

The Interest Rate Elasticity of Investment: Micro Estimates and Macro Implications

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Abstract

I estimate the elasticity of investment to interest rates using cross-sectional variation and high-frequency monetary shocks. My estimates imply that a shock that lowers real interest rates by 1 p.p. increases capital demand by 2%. This implies a significant effect of interest rates on investment but is much smaller than prominent estimates of the interest rate elasticity derived from the investment response to tax policy changes. In a quantitative model with heterogeneous firms, I show that the impulse response I estimate is a powerful tool for discriminating between models with different frictions. The evidence favors models with financing constraints: models with real adjustment costs generate nearly identical responses to interest rate and tax policy, while models with financing constraints generate substantially larger responses to tax policy, as in the data. Calibrating a New Keynesian model to these estimates, rather than to tax-based elasticities, halves the power of monetary policy.

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The discrepancy between theory and empirical work is perhaps nowhere in macroeconomics so obvious as in the case of the aggregate investment function.

Olivier Blanchard, 1986

1 Introduction

Investment is one of the core macroeconomic aggregates studied by researchers and policy-makers and is crucial for understanding business cycles and the effects of interest rate policy. Investment can be a powerful amplifier of monetary policy if higher investment induced by lower interest rates translates into higher labor and capital income and, ultimately, consumption and output. As a result, understanding investment's response to interest rates is vital for understanding the overall effect of monetary policy. Standard models of business cycles predict very large responses to even relatively modest changes in interest rates.

Why is investment so responsive to changes in interest rates? The characteristic of capital goods that drives these large responses is that capital is durable. Consider a firm that purchases an additional unit of capital and uses that capital for production immediately upon purchase. In the next period, the firm retains $1-\delta$ of the value of that unit of capital, where δ represents the rate of economic depreciation. As a result, the firm's return on that investment includes not just the marginal production generated by that capital, but also the value of the capital that remains after depreciation. Thus, the allocative price of capital goods is not the price the firm pays for the additional unit of capital, but rather the *user cost of capital*, or the price of purchasing the capital in period t instead of purchasing in period $t + 1$. In neoclassical models, the interest rate affects investment because it is an input into the user cost: the interest rate functions as the opportunity cost of investing in period t . Even in models where firms cannot resell their non-depreciated capital, the durability of capital goods gives firms a powerful incentive to accelerate investment from periods where the user cost is relatively high to periods where the user cost is relatively low.

Despite the importance of investment to our understanding of business cycles, estimates of the investment response to interest rates vary widely. Because monetary policy is endogenous and investment is the most cyclical component of GDP, it is difficult to separate the effect of interest rates from other factors that vary over the business cycle. Traditional VAR methods require potentially implausible identifying assumptions, and high-frequency identification methods offer relatively little power to detect precise effects using time-series methods (Nakamura and Steinsson, 2018a). In addition, investment responses to interest rate changes iden-

tified through time-series methods reflect a combination of the partial equilibrium response to changes in the cost of capital and the response to general equilibrium effects that are also triggered when interest rates change, such as shifts in aggregate demand and Fed information effects. Disentangling these effects is essential for ensuring macroeconomic models reflect the underlying mechanisms that drive firms' investment choices and accurately computing the effects of counterfactual policies.

A recent literature in macroeconomics has used the well-identified, partial equilibrium consumption response to stimulus checks (the marginal propensity to consume, or MPC) to discipline general equilibrium models. Just as the investment response to interest rates reflects a combination of PE and GE forces, the effects of fiscal stimulus checks depend on both the MPC and the strength of general equilibrium feedback of increased demand on labor incomes. While estimates of the MPC alone cannot provide a comprehensive estimate of the aggregate effects of fiscal stimulus policies, they have proven crucial to ensuring macroeconomic models are quantitatively realistic and accurately reflect how households respond to shocks. In contrast, the literature estimating firm-level partial equilibrium investment moments and using those moments to discipline macroeconomic models of the business cycle is less well-developed. In addition, the partial equilibrium response of investment to interest rates is a key input for analyzing a wide range of macroeconomic policies. For instance, evaluating how tax changes affect output requires understanding the extent to which higher deficits crowd out investment. In such applications, isolating the direct response of investment to interest rates, abstracting from aggregate demand or information effects that are specific to monetary policy shocks, is crucial.

In this project, I use the fact that the durability of capital is precisely the characteristic that gives firms an incentive to accelerate investment in response to transitory interest rate changes to derive a novel estimator of the interest rate elasticity of investment. While it is standard to model a single type of capital good, firms use different types of capital goods with differing depreciation rates. This implies that firms that use different types of capital have differing incentives to increase their capital purchases in response to the exact same change in interest rates. I show that in a simplified, partial equilibrium model of investment, comparing the investment response across firms with different depreciation rates directly identifies the parameter that determines the responsiveness of investment to changes in the cost of capital and, thus, interest rates.

I then implement my estimation strategy in quarterly, firm-level data. I combine cross-sectional variation in depreciation rates across industries with time-series variation in interest rates from high-frequency shocks to monetary policy to identify the investment response to changes in interest rates. Because my empirical strategy includes sector-by-time fixed effects,

the regression differences out general equilibrium factors and isolates the partial equilibrium response of investment through changes in the cost of capital.¹ Consistent with neoclassical theory, firms with longer-lived capital goods increase their investment substantially more than firms with shorter-lived capital after a surprise monetary easing. Through the lens of the simplified model, the estimates imply substantial effects of interest rates on investment and capital: in response to an expansionary shock that lowers real interest rates by 1 p.p., aggregate capital demand increases by around 2% after one year. This effect persists until about two years after the monetary shock when the capital response begins to decline towards zero.

While there are limited partial equilibrium estimates of the investment response to interest rates, recent work (Koby and Wolf, 2020) has used estimates of the investment response to transitory changes in corporate taxes to discipline the investment response to interest rates. In particular, because corporate taxes and interest rates both affect investment by changing the cost of capital, the investment response to tax changes (which I refer to as the “tax rate elasticity”) is informative about the investment response to changes in interest rates (which I refer to as the “interest rate elasticity”). In standard models, the interest rate elasticity is simply a rescaling of the tax rate elasticity.

However, I show that my estimates of the interest rate elasticity are substantially smaller than the interest rate elasticities implied by prominent estimates of the tax rate elasticity (Zwick and Mahon, 2017). The *annual* implied interest rate elasticities estimated from tax rate changes are about 4.5 times larger than the interest rate elasticities estimated directly from monetary shocks. Thus, my estimates suggest that using the tax rate elasticity to proxy for the investment response to interest rates will likely result in a substantial overestimate of the interest rate elasticity. In addition, the estimates suggest that traditional, neoclassical models where transitory changes in interest rates and tax rates have similar effects on investment are misspecified.

I then bring my empirical estimates to a quantitative model where heterogeneous firms make investment choices to highlight how the empirical moments estimated in this project can discipline the quantitative magnitude of the investment response to interest rates and discriminate between models where firms face different types of frictions. I use the empirical estimates to calibrate two separate models of investment: one in which firms face convex and fixed adjustment costs and another where firms face financial frictions that make it costly to borrow to fund investment. I target the estimated empirical impulse response from monetary shocks to calibrate the strength of both the adjustment costs and the financial frictions. Consistent with

¹In this paper, I define industries (i) as disaggregated subsets of sectors (s) (e.g. $i \subseteq s$). For example, I define “Durable Manufacturing” as a sector and “Equipment Manufacturing” as an industry that is part of the “Durable Manufacturing” sector.

the modest interest rate elasticities estimated in the data, the models require very tight financial frictions or strong adjustment costs to match the data. However, the quantitative results also show that financial constraints alone can generate investment elasticities that match the data without appealing to arbitrary adjustment costs.

I then use the tax rate elasticity from Zwick and Mahon (2017) as an untargeted moment to discriminate between the two types of models. The model with strong real adjustment costs cannot simultaneously generate modest responses to interest rate policy and larger responses to tax policy changes. In contrast, the model without adjustment costs but with strong financial frictions generates responses to tax rate changes that are more than twice as large as the response to interest rate changes, substantially shrinking the gap between the two estimates. Why can only the model with financial frictions generate substantial differences between tax-rate elasticity and interest-rate elasticity? In neoclassical models, taxes and interest rates both affect investment by changing the cost of capital. Adjustment costs slow the investment response to cost of capital shocks, but have similar effects regardless of the policy instrument that changes the cost of capital. Thus, investment elasticities to tax rates and interest rates will be similar when adjustment costs are important, just as in frictionless models.

In contrast, in the presence of financial frictions, tax rates have a second important effect. By changing the immediate tax bill the firm owes to the government, tax policy changes the amount of free cash flow available to the firm to fund investment. This cash-flow effect is irrelevant in both frictionless models and models with real adjustment costs: the firm will costlessly borrow and invest optimally. However, this additional cash-flow effect can magnify the investment response when financial frictions constrain firm borrowing or investment. In the estimated model with financial frictions, firms have a large marginal propensity to invest (MPI) of 0.61, implying that every dollar the government transfers to the firm generates 61 cents of investment (Winberry et al., 2025). Because changes in interest rates do not generate immediate cash flows of the same magnitude in the presence of financial frictions, tax rate elasticities are larger than interest rate elasticities.² Consequently, my estimates suggest that monetary policy is much less effective at stimulating investment spending than tax policy, which has important implications for countercyclical policy design.

To validate the theoretical mechanism, I return to the data and estimate heterogeneous interest rate elasticities by firm characteristics. I estimate whether firms more likely to face tighter financial constraints respond differently to those less likely to be financially constrained after interest rate shocks. I focus on firms' interest expenses to proxy for how close firms are to

²In the presence of debt financing, monetary policy does generate cash flow by reducing interest expenses (Gürkaynak, Karasoy-Can and Lee, 2022). However, these cash flow benefits are substantially smaller and less immediate than the cash flow effects of tax policy, as discussed in Section 6.

interest-coverage covenants, which are the most common type of debt covenant (Greenwald, 2019).

I find substantial differences in interest rate elasticities by firms' levels of interest expenses. Firms with high interest expenses exhibit smaller elasticities, while firms with low interest expenses have significantly larger responses than implied by the baseline estimates. In fact, the empirical monetary response of low interest expense firms is very similar to the response of firms to tax policy in the model, providing additional evidence that financial frictions are responsible for a significant share of the gap between the interest rate and tax rate elasticities.³ In contrast, I find no significant evidence of heterogeneous responses by several other characteristics hypothesized to affect the investment response to interest rates: firm leverage, liquidity, and fixed costs. Overall, I interpret these results as providing additional evidence for the core mechanism in the model: financial frictions that limit a firm's external financing play an essential role in reducing the investment response to interest rate changes. These results also imply that tax rate elasticities will be larger than interest rate elasticities because tax policy shocks also free up cash flow within the firm that can be used for investment.

What do these results imply for the aggregate effects of monetary policy? In the final part of the paper, I calibrate a tractable, heterogeneous-agent New Keynesian model with capital to the PE interest rate elasticities to examine what the results imply in general equilibrium. I find that calibrating to the empirical interest rate elasticities, rather than the existing estimates of tax rate elasticities, has large effects on the GE effects of monetary policy. Overall, the immediate effect of an interest rate cut on investment is about 70% weaker when calibrating to the interest rate elasticities, while the effects on output and consumption shrink by 60% and 44%, respectively. Thus, my results suggest that the power of monetary policy is significantly weaker once one accounts for the more limited response of investment estimated in this paper.

Overall, my empirical strategy provides a novel estimate of the effect of interest rates on investment. Using the quantitative models, I show that this estimate is a useful "portable moment" (Nakamura and Steinsson, 2018b) that can be used to discipline and discriminate between competing theoretical models. In particular, when combined with recent estimates of the investment response to tax policy, I show that my estimates provide evidence in favor of models where financing constraints reduce the investment response to interest rates, as opposed to models with strong real adjustment costs. My results also show that monetary policy has substantially weaker effects on investment, output, and consumption, once models are calibrated to the smaller partial equilibrium investment elasticities.

³However, in both the model and the heterogeneity tests, a substantial gap remains unexplained, suggesting that financial frictions cannot explain the entire difference between the investment response to the two policy shocks.

1.1 Literature Review

My research is related to three main strands of the literature. In public finance and macroeconomics, a substantial literature has focused on understanding the nature of firms' investment decisions (Hayashi, 1982; Shapiro, 1986; Caballero, 1994; Cummins, Hassett and Hubbard, 1994; Caballero, Engel and Haltiwanger, 1995; Chirinko, Fazzari and Meyer, 1999; Hassett and Hubbard, 2002; House and Shapiro, 2008; Zwick and Mahon, 2017). I add to this literature by deriving a new quasi-experimental estimate of the investment response to changes in the cost of capital from interest rate shocks. I also show that financing constraints can provide a realistic alternative to real adjustment costs in reducing the large investment elasticities from interest rate changes implied by frictionless models while remaining consistent with other features of the data.

A second body of literature has studied the transmission of monetary policy through firms, which has largely focused on how credit constraints or the firm lifecycle affect the investment response to changes in interest rates (Gertler and Gilchrist, 1994; Jeenas, 2026; Ottonello and Winberry, 2020; Bahaj et al., 2022; Cloyne et al., 2023; Caggese and Pérez-Orive, 2022; Döttling and Ratnovski, 2023; Jeenas and Lagos, 2024; Fang and Zhou, 2025; Gnewuch and Zhang, 2025; Modi, 2024; Selgrad and Siani, 2025; Drechsel et al., 2026). Relative to this work, this project provides a direct estimate of the neoclassical investment elasticity implied by theoretical models. The most closely related paper is contemporaneous work by Best, Born and Menkhoff (2026), which uses a high-quality German survey to directly ask firms how changes in interest rates would affect their investment behavior. In contrast, my work uses quasi-experimental techniques in the full sample of U.S. publicly traded firms to estimate a similar parameter. Despite these methodological differences, our estimates are similar: Best, Born and Menkhoff (2026) estimates a one-year investment semi-elasticity of -6, and this paper estimates a one-year semi-elasticity of -8. As a result, both results imply much smaller investment responses to interest rate policy than those identified from tax policy changes. Thus, the results from both papers are highly complementary.

In subsequent work, Selgrad and Siani (2025) study how firms adjust their investment plans in response to monetary policy shocks. Interestingly, and though it is not the focus of either paper, both Best, Born and Menkhoff (2026) and Selgrad and Siani (2025) provide supportive evidence of the mechanisms first documented in this paper: Best, Born and Menkhoff (2026) elicit higher investment responses from managers of firms in lower depreciation industries, while Selgrad and Siani (2025) show that firms in lower depreciation industries revise their investment plans more strongly in response to interest rate policy. Thus, both papers provide additional evidence for the underlying user-cost channel that is the focus of this paper.

Krusell, Thürwächter and Weiss (2026) and Cao et al. (2025) both use time-series style

identification and microdata to estimate the response of investment to interest rates. In contrast, my difference-in-differences strategy adds power by combining cross-sectional variation with high-frequency monetary shocks. My identification strategy also directly identifies the effect of changes in the cost of capital, separating those forces from other general equilibrium factors. I also show that a particular type of financial friction, an external financing constraint, can help explain both the lower investment elasticities estimated in this project and the higher elasticities estimated from changes in tax policy.

Two recent papers document sectoral heterogeneity in investment response to interest rates (Howes, 2025; Durante, Ferrando and Vermeulen, 2022). In contrast, this project includes sector-time fixed effects and thus carefully controls for these sectoral differences, and instead looks at differences within sectors in the durability of the investment goods used by each industry in production. Bloesch and Weber (2026) argue that secular changes in the composition of investment have substantially weakened the investment channel of monetary policy. Koby and Wolf (2020) combine theoretical models with quasi-experimental results from the literature on bonus depreciation policies to make broader inferences about aggregate and firm-level investment behavior. In contrast, I use similar identifying variation to directly estimate the effects of monetary policy on firms while carefully accounting for the impact of bonus depreciation.

Finally, my work contributes to the literature on the monetary transmission mechanism (Romer and Romer, 2004; Christiano, Eichenbaum and Evans, 2005; Smets and Wouters, 2007; Kaplan, Moll and Violante, 2018; Auclert, Rognlie and Straub, 2020; Winberry et al., 2025). I provide new empirical evidence on monetary transmission through investment. I provide a new portable empirical target that can be used to discipline business-cycle models with capital, and show that models correctly calibrated to these estimates imply that the aggregate effects of monetary policy are about 50% weaker than when calibrating to existing estimates. In addition, Winberry et al. (2025) argue that the MPI is a crucial parameter for business cycle models with firm and worker heterogeneity, but find very little quasi-experimental evidence on the magnitude of the parameter. My estimates, interpreted through the lens of the quantitative model, imply an MPI of 0.61. Kaplan, Moll and Violante (2018) and Auclert, Rognlie and Straub (2020) emphasize how estimates of the MPC can be used to discipline the consumption response to monetary policy. In this paper, I show how my estimate of the interest rate elasticity can be used to discipline the investment response to monetary policy.

The rest of the paper is organized as follows. Section 2 uses a standard model of investment to provide intuition for the identification strategy and explicitly derives the estimator used in this paper. Section 3 presents the main data sources used in the paper. Section 4 presents the baseline investment results. Sections 5 and 6 develop the quantitative model with heterogeneous firms and show that the empirical results from Section 4 can be used to discriminate

between modern, quantitative models featuring different firm-level frictions. In Section 7, I return to the data and provide evidence of heterogeneous responses that validates the core mechanism from the quantitative model. Finally, Section 8 discusses how the partial equilibrium elasticity affects the aggregate effect of monetary policy in general equilibrium and concludes.

2 The Interest Rate Elasticity in a Neoclassical Model

2.1 Benchmark Model

In this section, I present a neoclassical model of firm investment to highlight how investment responds to changes in interest rates and derive a new empirical estimator of the interest rate elasticity of investment. This model highlights a core feature of all neoclassical models of capital accumulation: the durability of capital makes firms' investment choices very sensitive to changes in interest rates. I begin with a one-factor model, where firms produce using capital and are subject to heterogeneous idiosyncratic productivity shocks ($z_{f,t}$). Firms also differ in the rate at which their capital depreciates (δ_f). Firms choose their capital in each period to maximize a discounted stream of dividends:

$$\max_{\{K_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \left(\prod_{s=1}^t (1 + R_s + \Theta) \right)^{-1} (z_{f,t} K_{t-1}^{\alpha} - I_t - \psi(K_{t-1}, K_t))$$

Here, the firm's production function takes as its inputs the firm's idiosyncratic productivity and last period's capital.⁴ The firm's remitted dividends are then equal to the value of the firm's production, minus its investment choice and any adjustment costs the firm pays. The firm's budget constraint is: $K_t = I_t + (1 - \delta_f)K_{t-1}$. Finally, Θ represents a *discount-rate wedge*, which measures the additional return firms require to make the owner indifferent between allocating a marginal dollar to real investment and earning the safe, one-period interest rate (R). This captures the fact that corporate investment is risky, and firms report discounting cash flows at rates that are higher than the safe interest rate (Gormsen and Huber, 2025). Assume the adjustment cost function is differentiable, then the Euler equation that characterizes the firm's choice of capital each period is given by:

$$1 + \psi_2(K_{t-1}, K_t) = \frac{1}{1 + R_{t+1} + \Theta} \mathbb{E}_t [\alpha z_{f,t+1} K_t^{\alpha-1} + (1 - \delta_f) - \psi_1(K_t, K_{t+1})] \quad (1)$$

Equation 1 states that the firm invests until the marginal costs of additional capital are

⁴Thus, the model implicitly includes a "time-to-build" friction.

equal to the marginal benefits. The left-hand side of Equation 1 represents the marginal costs of purchasing an extra unit of capital: the price of capital (here, normalized to one), plus the marginal adjustment cost for installing that capital. The right-hand side of Equation 1 represents the marginal benefits of additional capital: the marginal product of capital from production in period $t + 1$, plus the extra value to the firm of the non-depreciated capital stock,⁵ and the reduction in future adjustment costs the firm pays from investing today. These benefits are then discounted using the firm's discount rate: $R_{t+1} + \Theta$.

Equation 1 provides intuition for how depreciation rates affect a firm's investment decisions. Firms with more durable capital retain more of the value of that capital after it is used for production. In models where firms can resell that capital to the market, that value is captured by the capital's price. However, even in models where firms cannot resell capital, the durability of capital gives firms strong incentives to respond to transitory interest rate changes because durability allows firms to retime investment from periods when R is relatively high towards periods when R is relatively low, and thus to periods where firms are more patient.

How strongly should firms react to a change in interest rates? Consider a simplified version of the model without adjustment costs. Then, the firm's first-order condition for capital is given by:

$$\alpha \mathbb{E}_t[z_{f,t+1}]K_t^{\alpha-1} = R_{t+1} + \delta_f + \Theta = \mu_{f,t} \quad (2)$$

This is the well-known expression that the firm invests until the marginal product of capital is equal to $\mu_{f,t}$, the *user cost of capital* (Hall and Jorgenson, 1967). In this variant of the model, the magnitude with which capital responds to changes in interest rates is given by the following expression, where $\Psi = \frac{1}{\alpha-1}$

$$\frac{\partial \log K_t}{\partial R_{t+1}} = \frac{\Psi}{\delta_f + R_{t+1} + \Theta} \quad (3)$$

The strength of the investment response to changes in real interest rates is determined by Ψ , which is the user cost elasticity of investment.⁶ Note that this is the partial equilibrium response, as all other prices and variables in the model are held fixed. In general equilibrium, firms' investment responses could be larger or smaller than in the partial-equilibrium benchmark. For example, if expansionary monetary policy increases demand for a firm's product, that would increase demand for capital used by that firm in production, and thus magnify the response. However, if expansionary policy provides negative information about the state of the economy to the market, as argued by the literature on the Fed Information Effect (Nakamura

⁵In this model, this is equal to capital's resale value.

⁶We define Ψ as the user cost elasticity because $\Psi = \frac{\partial \log K_t}{\partial \log \mu_{f,t}} = \frac{1}{\alpha-1}$.

and Steinsson, 2018a), this would lead firms to reduce their forecast of future demand growth, and thus reduce demand for capital. This simplified model clarifies, however, that in partial equilibrium, interest rates affect investment and capital accumulation through their effect on the user cost of capital.

In this simplified model, the user cost elasticity (Ψ) is driven by the curvature of the firm's revenue function.⁷ In a standard parameterization (Winberry, 2021), which excludes a discount-rate wedge and has $\alpha = 0.7$, $R_{t+1} = 0.04$, and $\delta = 0.10$, capital is very responsive to changes in interest rates: in response to a 1 percentage point increase in the real interest rate, the capital stock will fall by more than 20%. It is also clear from Equation 3 that the magnitude of the response of capital to any given interest rate shock is decreasing in the firm's depreciation rate.

2.2 Empirical Estimator

Equation 3 highlights that in models without adjustment costs, the critical parameter governing the investment response to interest rate changes is the user cost elasticity of investment. Equation 3 also shows that the investment response to interest rates will differ with the firm's rate of economic depreciation. I can now derive the estimator in this paper by showing that comparing the investment response for firms with different depreciation rates can directly identify Ψ and thus the interest rate elasticity. Consider two firms, one with a relatively low rate of economic depreciation ($f = l$) and one with a higher rate of economic depreciation ($f = h$). Taking log differences of the first-order conditions for capital across firms and across the period before and after a change in interest rates leads to the following regression specification:

$$\Delta \log(k_{l,t}) - \Delta \log(k_{h,t}) = \Psi \{ \Delta \log(\mu_{l,t}) - \Delta \log(\mu_{h,t}) \} - \Psi \{ \Delta \log(\mathbb{E}_t[z_{l,t+1}]) - \Delta \log(\mathbb{E}_t[z_{h,t+1}]) \} \quad (4)$$

Equation 4 specifies that comparing differences in the net investment rate after an interest rate shock for firms with different depreciation rates directly identifies the user cost elasticity of investment. Then, given a baseline user cost, this implies an interest-rate semi-elasticity of $\frac{\Psi}{\mu}$. In practice, I estimate a variant of Equation 4 to maintain comparability with previous work. However, rather than putting the period-by-period log change in the user cost on the right-hand side, I estimate the differential response of net investment to an identified monetary shock and then rescale those differences by the differential proportional response of the user cost across firms. This flexibly allows for a dynamic response where changes in the capital

⁷This can be microfounded either through decreasing returns to scale or through the firm's demand elasticity.

stock do not have to exactly coincide with the dynamics of the changes in interest rates.

This regression equation was derived after abstracting from adjustment costs. Equation 3 states that the curvature of the revenue function determines the total change in the capital stock, because that parameter determines how much individual firms want to scale up in response to changes in their costs. In reality, the timing and magnitude of the response are likely determined by both the degree to which firms want to scale up, as well as the degree to which factors of production are substitutable with capital and other frictions faced by firms. In Section 6, I bring the estimated impulse response to a structural model where all of these forces together determine the magnitude and the timing of the response. As a result, in the structural model, the regression coefficients I compute do not identify a single structural parameter, but rather reflect a combination of structural parameters and investment frictions.

2.3 Identification and Diagnostics

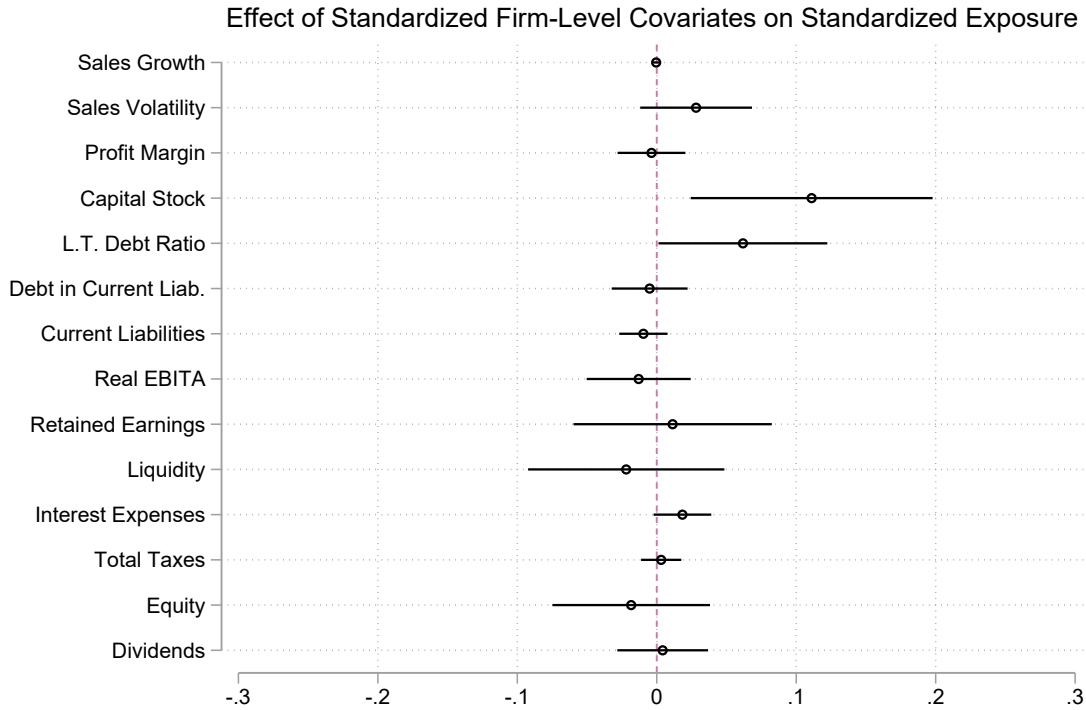
Equation 4 also clarifies the identification assumptions underlying the differences-in-differences empirical strategy. If changes in productivity are unobserved, then they will enter the error term in the regression. Define $\epsilon_t = \Delta \log(\mathbb{E}_t[z_{l,t+1}]) - \Delta \log(\mathbb{E}_t[z_{h,t+1}])$. Then, the key assumption for identification is:

$$\mathbb{E}[\epsilon_t \{\Delta \log(\mu_{l,t}) - \Delta \log(\mu_{h,t})\}] = 0 \quad (5)$$

Through the lens of the benchmark model, changes in $\mathbb{E}_t[z_{f,t+1}]$ also influence a firm's investment choices. If $\mathbb{E}_t[z_{f,t+1}]$ is unobserved, and changes in $\mathbb{E}_t[z_{f,t+1}]$ are correlated with changes in the user cost, this would lead to biased estimates of Ψ . Because I use high-frequency monetary shocks and differences in depreciation rates to generate changes in the user cost, the broader identification argument relevant to this project is that, after residualizing on sector-by-time fixed effects, productivity changes, as well as other factors outside the simplified model that affect investment, including shifts in product demand and any other factors that drive investment after monetary shocks, are not systematically correlated with depreciation rates.

While this assumption cannot be tested empirically, Figure 1 shows the coefficients from a multivariate regression of time-varying, firm-level characteristics hypothesized to affect investment on the economic depreciation rate after residualizing on sector-by-year fixed effects and the treatment measure capturing the supply of capital goods. Nearly all coefficients are close to zero and statistically insignificant, implying that underlying firm fundamentals are similar across firms with differing depreciation rates. The one large and statistically significant coefficient is the total size of the capital stock. Firms with lower depreciation rates own more capital. However, this difference is consistent with our benchmark model. Equation 2

Figure 1: Correlation of Firm-Level Outcomes with Standardized Exposure



Notes: This figure plots the coefficients of a multivariate regression of standardized, firm-level characteristics on standardized exposure. 95% confidence intervals are shown, with standard errors two-way clustered at the industry and date levels.

states that firms should invest until the point where the marginal product of capital equals the user cost, including the depreciation rate. Taking a derivative with respect to the depreciation rate suggests that, holding all else equal, lower-depreciation firms should have more capital because they face lower user costs of capital.

The second important, non-zero coefficient is the coefficient on the firm's long-term debt ratio. Though only marginally statistically significant, the coefficient implies that lower depreciation firms finance more of their assets with long-term debt, consistent with maturity matching. To investigate whether this affects the interpretation of the main results or sheds light on the underlying mechanisms, I investigate the effect of interest rate shocks on firm debt financing. I also study the investment response to interest rate shocks that affect different parts of the yield curve. As a result, I can test whether the investment response is larger for interest rate shocks that affect longer-term yields, which have more relevance for corporate bond contracts. Finally, and reassuringly, in the Appendix, I show that my results are quite insensitive to the inclusion of firm- or industry-level controls and to the addition of additional controls, suggesting that the baseline identification strategy creates a relatively balanced set of implicit

treatment and control firms and industries.

Throughout this paper, I focus on how interest rates affect capital accumulation and investment through changes in the demand for capital goods. However, firms can also be exposed to the investment channel of monetary policy through the supply of capital if they produce capital goods. Relative to the demand channel, identifying the investment channel of monetary policy through the supply of capital goods is challenging because a substantial share of investment goods are imported (Bloesch and Weber, 2026), and the production of investment goods is concentrated in a relatively small number of industries (vom Lehn and Winberry, 2022). As a result, while the results in this paper focus primarily on exposure through the demand for capital goods, throughout the paper we define a second treatment which captures exposure through the supply of capital goods, based on data from vom Lehn and Winberry (2022), and control for both measures in any regressions.

While Figure 1 provides evidence that firms facing different depreciation rates are balanced along observable characteristics, a potential concern is that there is unobserved heterogeneity correlated with depreciation rates that affects how firms respond to interest rates. In particular, if unobserved adjustment costs or financial frictions vary across firms in ways that are correlated with depreciation rates, this can bias estimates of the user cost elasticity. However, as I show in Section 6, the structural model provides a simple statistic that can be used to investigate the extent to which these (normally) unobserved investment frictions bias the results in practice. In particular, I show that the strength of these investment frictions is a key determinant of the variance of the firm-size distribution: firms and industries that face larger adjustment costs or financial frictions respond less to idiosyncratic shocks, which leads to more compressed firm-size distributions within an industry. Thus, I can use the variance of the firm-size distribution across industries to assess and correct for the degree to which investment frictions vary across industries.

In practice, while Figure 1 shows that average firm size is strongly correlated with depreciation rates, Appendix Figure A1 shows that the variance of the firm-size distribution does not covary strongly with industry-level depreciation rates. Panels (C) and (D) show, using binned scatter plots, that there is no relationship between the depreciation rate and variance of the firm-size distribution at the industry level. The point estimates of the regression are very close to zero and statistically insignificant. As a result, we find evidence that heterogeneity in investment frictions across firms is unlikely to bias our estimates of the user cost elasticity.

Most importantly, the identification assumption underlying my estimates is very similar to that used in the literature estimating the user cost elasticity from the investment response to bonus depreciation policies in the United States (House and Shapiro, 2008; Zwick and Mahon, 2017; Curtis et al., 2023). This literature compares investment across industries using

capital goods with different tax recovery periods, reflecting differences in asset durability, and therefore differential exposure to policies that accelerate depreciation deductions.⁸ In Section 4, I compare my estimates with the estimates from this literature. Because the two empirical designs rely on a similar set of identifying assumptions, the two sets of estimates are directly comparable, reducing the scope for unobserved heterogeneity to explain differences between the two sets of estimates.

3 Data

I implement my estimation strategy in quarterly firm- and capital-good level data.

Industry-Level Capital Stocks: The primary source of identification in the paper comes from industry-level differences in the type of capital used in production, based on data from the Bureau of Economic Analysis (BEA). The BEA provides data on the current-cost net capital stock of private nonresidential fixed assets across various industries and a crosswalk between the BEA industries and the corresponding NAICS industries. In addition, BEA provides implied depreciation rates at the capital asset-industry-year level. I use these two data sources to measure average industry-level depreciation, using a weighted average of the implied depreciation rates with the asset type’s share in the total capital stock used as the weighting variable. I use data from 1990, four years before the main sample period, to create the industry-level depreciation rate. Appendix Table A1 shows the economic depreciation rate for the highest and lowest depreciation industries after residualizing the depreciation rates on sector fixed effects.

Firm-Level Data: I use quarterly data from Compustat (provided through WRDS) to estimate the effect of monetary policy shocks across industries. The primary outcome variable of interest is net capital investment: the log change in the firm’s capital stock between $t - 1$ and $t + h$. I construct the firm’s capital stock by summing net property, plant, and equipment, as in Ottonello and Winberry (2020). Within each quarter, I winsorize the value of the dependent variable at the 1% level. I follow standard practice and exclude firms in finance and utilities. Table 1 presents summary statistics of the main firm-level variables.

Monetary-Policy Shocks: To identify the effect of monetary policy, I use replication data from Gürkaynak, Karasoy-Can and Lee (2022) to construct monetary-policy shocks using high-frequency changes in asset prices in 30-minute windows around monetary-policy announcements. I use the first principal component of the change in the same five asset prices as Gürkaynak, Sack

⁸House and Shapiro (2008) look at variation across capital goods, rather than variation across industries.

Table 1: Summary Statistics for the Compustat Sample

	N	Mean	Median	SD	P10	P90
D.Log Capital	263,450	0.00	-0.00	0.08	-0.06	0.07
1-Year D.Log Capital	248,950	0.02	-0.00	0.28	-0.21	0.30
2-Year D.Log Capital	230,312	0.05	0.01	0.47	-0.38	0.56
1-Year Cumulative Capex / L.Capital	197,108	0.23	0.15	0.26	0.04	0.50
2-Year Cumulative Capex / L.Capital	152,014	0.48	0.32	0.54	0.08	1.03
D.Log Cash and Short-Term Assets	259,926	0.02	0.00	0.67	-0.61	0.66
Real Sales Growth	263,554	0.01	0.01	0.22	-0.21	0.23
Log Real Assets (USD Millions)	263,554	5.36	5.37	2.33	2.34	8.37
Share Current Assets in Total Assets	263,554	0.52	0.53	0.25	0.17	0.85

Notes: This table presents summary statistics for the sample of U.S. firms from Compustat.

and Swanson (2005) and Nakamura and Steinsson (2018a) to measure the policy changes.⁹ I sum announcement-level shocks within each quarter to aggregate shocks to the quarterly level. Appendix Figure A2 shows the time series of the high-frequency monetary shocks during my sample period.

Investment-Network Data: To construct measures of exposure to monetary policy through the production of capital goods, I use data on industry-level production of investment goods and industry crosswalks from vom Lehn and Winberry (2022). I create an indicator variable for whether or not industries supply capital goods based on the share of total capital goods they produce in 1990.

Aggregate Capital Investment and Prices: I use data from BEA to measure nominal gross investment and capital prices for 36 investment goods from 1994 to 2017. I merge these measures to tax-depreciation schedules from House and Shapiro (2008) and Zwick and Mahon (2017) to construct capital-good-level measures of exposure to bonus depreciation. For analysis using aggregate BEA data, the primary variables of interest are gross investment and changes in asset-specific price indices for capital goods. Table 2 shows summary statistics for the primary variables from the BEA data.

⁹The five asset prices capture expected changes in the federal funds rate over the current month and after the next scheduled FOMC meeting, as well as changes in Eurodollar futures rates two, three, and four quarters ahead.

Table 2: Summary Statistics for Aggregate Quantities and Prices of Capital Purchases

	N	Mean	Median	SD	P10	P90
Gross Investment (s.a., ann., USD Mill.)	3,456	28,123.51	18,378.50	29,476.59	4,048.00	73,119.00
Capital Price Index	3,456	0.95	0.86	0.80	0.58	1.02
1-Year D.Log Investment	3,312	0.04	0.05	0.22	-0.19	0.26
1-Year D.Log Prices	3,312	0.01	0.02	0.05	-0.02	0.05

Notes: This table presents summary statistics for capital goods quantities and prices.

4 Empirical Results

4.1 Empirical Strategy

This section presents the baseline empirical results. I test whether lower interest rates lead to higher investment through partial equilibrium changes in the user cost of capital. I estimate the following regression using Jordà (2005)-style local projections.

$$\log k_{f,t+h} - \log k_{f,t-1} = \beta_h \epsilon_t \delta_i + \alpha_{f,h} + \alpha_{s \times t,h} + \Gamma' Z + \eta_h z_{i,t} + e_{f,t,h} \quad (6)$$

In this specification, h indexes the forecast horizon, $f \subseteq i \subseteq s$ indexes firms (f) that are part of industries (i) and sectors (s) and t indexes time (in quarters). The primary outcome variable, $\log k_{f,t+h} - \log k_{f,t-1}$ is the winsorized log change in the firm-level capital stock. In this regression, the primary variable of interest is $\epsilon_t \delta_i$, which represents the (standardized) industry-level depreciation rate interacted with ϵ_t , high-frequency monetary policy shocks.

The primary coefficients of interest are β_h , which represent the estimated differential impulse response of capital investment in industries with lower vs. higher rates of economic depreciation in response to monetary policy shocks. Because the identifying variation is at the time-by-industry level, I two-way cluster standard errors by industry and date. To rescale the β_h coefficients into user cost and interest rate elasticities that can be compared to theoretical models and the existing empirical literature, I apply Equations 3 and 4, derived using the benchmark model in Section 2.

The regressions include $\alpha_{f,h}$, firm fixed effects, and $\alpha_{s \times t,h}$ sector-by-time fixed effects. Thus, identification of the relative effects of monetary policy shocks comes from variation within a sector, and results are not driven by differential response of particular sectors (e.g. durable manufacturing, nondurable manufacturing, construction) to monetary policy. $Z_{f,t-1}$ is a vector of controls, including variation in firm-level financial constraints, proxied by the leverage ratio, and the interaction of the leverage ratio with the monetary policy shock (Ottonello and

Winberry, 2020).¹⁰

Other controls include firm-level (lagged) sales, assets, current assets, and a dummy indicating if the firm is in its fourth fiscal quarter, given the investment spikes in the fourth fiscal quarter documented by Xu and Zwick (2024). Finally, I include an interaction of the monetary policy shock with an indicator for whether a firm is in an industry that produces capital goods (vom Lehn and Winberry, 2022). This captures whether interest rates stimulate investment in companies that produce capital goods, such as machinery-manufacturing firms.

Given the tight link between the interest rate and bonus depreciation elasticity, a potential confound is that the results are driven by bonus depreciation rather than monetary policy. I address this concern in two ways: first, in both regressions, I only use shocks to monetary policy that can be identified through high-frequency variation, which should be orthogonal to bonus depreciation policies. Indeed, during the primary sample, the correlation between the quarterly monetary policy shock and the generosity of bonus depreciation is 0.09. Second, in all regressions, I include $z_{i,t}$: the Zwick and Mahon (2017) measure of bonus-depreciation generosity, which varies at the 4-digit NAICS industry by quarter level.¹¹

Finally, I use the unemployment rate to identify periods when the economy is in an expansion or contraction and interact both measures with the industry-level depreciation rate. Though the high-frequency identification should mitigate any concern about the correlation between monetary policy and the business cycle biasing the results, this ensures that the results are not driven by cyclical behavior in industries with more- or less-durable capital goods, conditional on sector-time fixed effects.

4.2 The Effect of Monetary Shocks on Interest Rates

First, I show that monetary shocks have a significant effect on interest rates. In neoclassical models, the short-run, one-period real interest rate is the relevant interest rate for firms' investment decisions.¹² As a result, I use the 1-month real interest rate as the baseline interest

¹⁰Allowing for young and middle-aged firms to respond differently to monetary shocks does not affect the results, ensuring that the age and life-cycle effects studied by Cloyne et al. (2023), Gnewuch and Zhang (2025), and Krusell, Thürwächter and Weiss (2026) do not affect the estimates.

¹¹Given the tight link between exposure to bonus depreciation and interest rates, the Zwick and Mahon (2017) measure of industry-level *exposure* to bonus depreciation (z_0) is highly correlated with the rate of economic depreciation. However, because tax depreciation is distinct from economic depreciation and because bonus depreciation generosity is uncorrelated with high-frequency monetary shocks, substantial variation exists to allow for separate identification of the effects of bonus depreciation and interest rate shocks.

¹²There is no distinction in either the benchmark model or in the structural model between shorter- and longer-term interest rates, as the expectations hypothesis holds in both models. Empirically, I investigate the response of investment to monetary shocks that move different parts of the yield curve, and find that the investment response is driven almost entirely by shocks to short rates.

rate.¹³

Figure 2 shows the impulse response of real one-month and one-year rates to a 25 basis point, expansionary monetary shock. Importantly, because I use movements in futures prices to construct the shocks, the shocks contain forward guidance about short-term rates in future quarters. I scale the monetary shock such that a 25-basis-point shock leads to an immediate 0.25 p.p. fall in the 3-month nominal Treasury rate. Following Ottonello and Winberry (2020), I weight the shocks by the number of days remaining in the quarter to reflect the fact that for earlier shocks firms have more time to respond.¹⁴ In response to this shock, real rates fall by around 25 b.p. in the shock quarter, implying close to full pass-through of nominal rates to real rates.¹⁵ The one-month rate continues to fall for another quarter after the shock, reflecting the fact that our measure captures forward guidance, before the real rate plateaus at approximately 0.6 p.p. lower than the baseline. The real rate then slowly reverts to its baseline level. Importantly, these shocks are much more persistent than conventional Taylor-Rule shocks in most quantitative macro calibrations.

One potential concern raised in the literature using high-frequency monetary shocks is that the monetary shocks are predictable, especially when aggregated to a quarterly frequency (Bauer and Swanson, 2023; Hubert de Fraisse et al., 2026). A natural interpretation put forward in both Bauer and Swanson (2023) and Hubert de Fraisse et al. (2026) is that the Federal Reserve may respond more quickly than markets to certain pieces of economic news. Alternatively, in relatively short samples like the period over which high-frequency shocks are available, correlations can arise spuriously for idiosyncratic reasons. As noted by both Bauer and Swanson (2023) and Hubert de Fraisse et al. (2026), this can confound the use of high-frequency shocks to identify the effects of monetary policy.

As shown in Figure 2, we find little evidence of pre-trends in real rates, which are the interest rates that matter for agents' decisions in the baseline model, in the quarters leading up to the monetary shock.¹⁶ In the quarter immediately preceding the shock, real rates are approximately unchanged relative to the preceding quarters. However, an advantage of the identification strategy used in this paper is that, rather than identifying the overall effect of monetary policy, it seeks to identify a precise channel through which monetary policy operates and uses an identification strategy that is explicitly designed to difference out both other channels of monetary policy and factors related to the business cycle and market conditions that

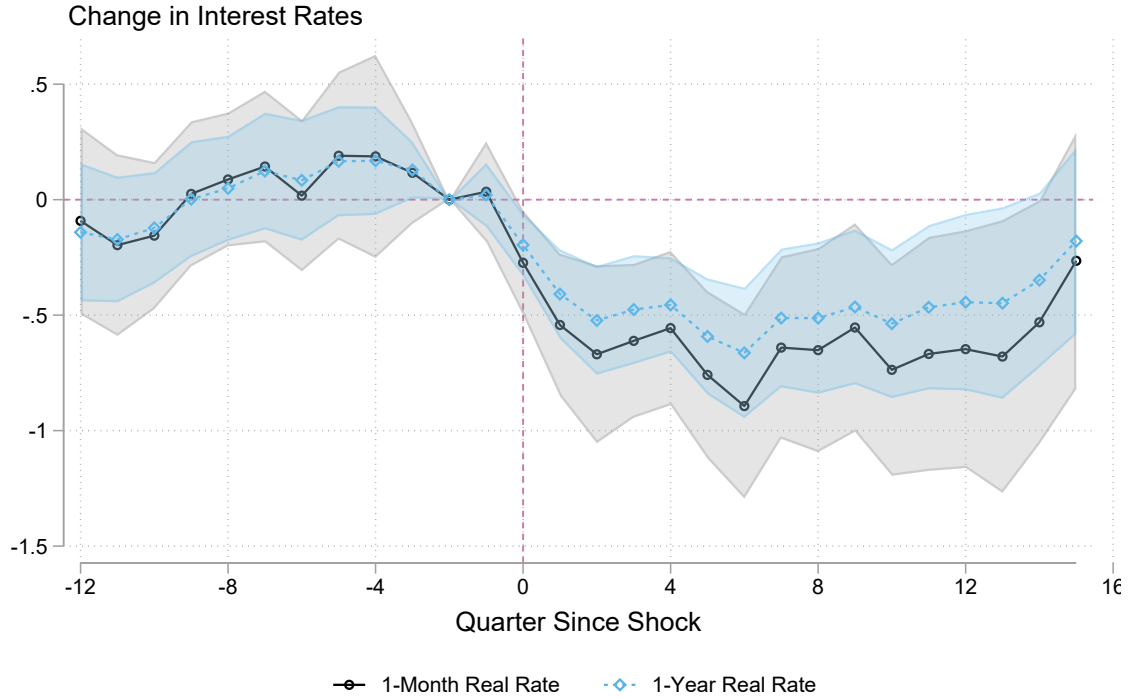
¹³The model and the empirical estimates are at a quarterly frequency. However, the Cleveland Fed does not publish a 3-month interest rate. Nevertheless, as shown in Figure 2, the shocks I use have very similar effects on both one-month (annualized) real rates and one-year real rates.

¹⁴However, results are very similar when the shocks are unweighted.

¹⁵This is consistent with the evidence presented in Nakamura and Steinsson (2018a).

¹⁶This is in contrast to the evidence for longer-term nominal rates presented by Bauer and Swanson (2023) and Hubert de Fraisse et al. (2026).

Figure 2: The Effect of Monetary Shocks on Short-Term Interest Rates



Notes: This figure plots the impulse response of one-month and one-year real treasury yields to a 25-basis point monetary policy shock.

could potentially drive Fed policy actions. In particular, the identification strategy used in this paper does not require an exogenous monetary policy shock; instead, Equation 5 requires that factors influencing a firm's investment decisions are uncorrelated with depreciation rates after residualizing on fixed effects and controls. Thus, this empirical strategy requires that monetary shocks move real interest rates, but does not require monetary shocks to be fully exogenous. In this sense, the empirical work in this paper is more similar to quasi-experimental work in public finance that studies the response of investment to tax policy both in that it targets a partial equilibrium response of investment, and in that it uses variation in responsiveness to particular channels of the policy across cross-sectional units for identification, rather than time-series variation in policy.¹⁷

A second advantage of this empirical strategy is that it provides a natural test of whether preexisting factors, such as news shocks that affect Federal Reserve behavior or business-cycle variation correlated with monetary shocks, can account for our results. If these factors are

¹⁷For example, the literature on bonus depreciation does not argue that bonus depreciation policies were exogenous. In fact, the early rounds of bonus depreciation in the United States were explicitly used for countercyclical stimulus.

significant confounds of our empirical strategy, they should lead to differences in investment between low- and high-depreciation firms in periods prior to the monetary shock. However, we find across all our baseline specifications and robustness tests that low- and high-depreciation firms were on very similar trends prior to shock quarters. Importantly, we find this without conditioning on any lags of the dependent variable and without strong controls for lags of the business cycle or for interest rate policy.¹⁸ Finally, in Appendix C, we show a range of robustness checks that allow us to rule out alternative stories that could explain our results. For example, we show that our results are quantitatively very similar when we drop shocks occurring during the major easing cycles of the early 2000s and during the Great Recession (Hubert de Fraisse et al., 2026). We also find similar quantitative results (and no evidence of pre-trends) when using narrative shocks or residuals from an AR(4) process to generate variation in real interest rates, which reflects the fact that our empirical strategy does not require exogenous monetary shocks for identification.¹⁹ This provides further evidence that our identification strategy successfully differences out other business-cycle factors or news shocks, isolating the partial-equilibrium investment channel through changes in the user cost.

4.3 The Effect of Interest Rates on Investment

Next, I show that movements in interest rates lead to significant changes in investment. Figure 3 shows the estimated β_h coefficients from Equation 6, which represent the interaction of the monetary policy shock with the industry-level depreciation rate.

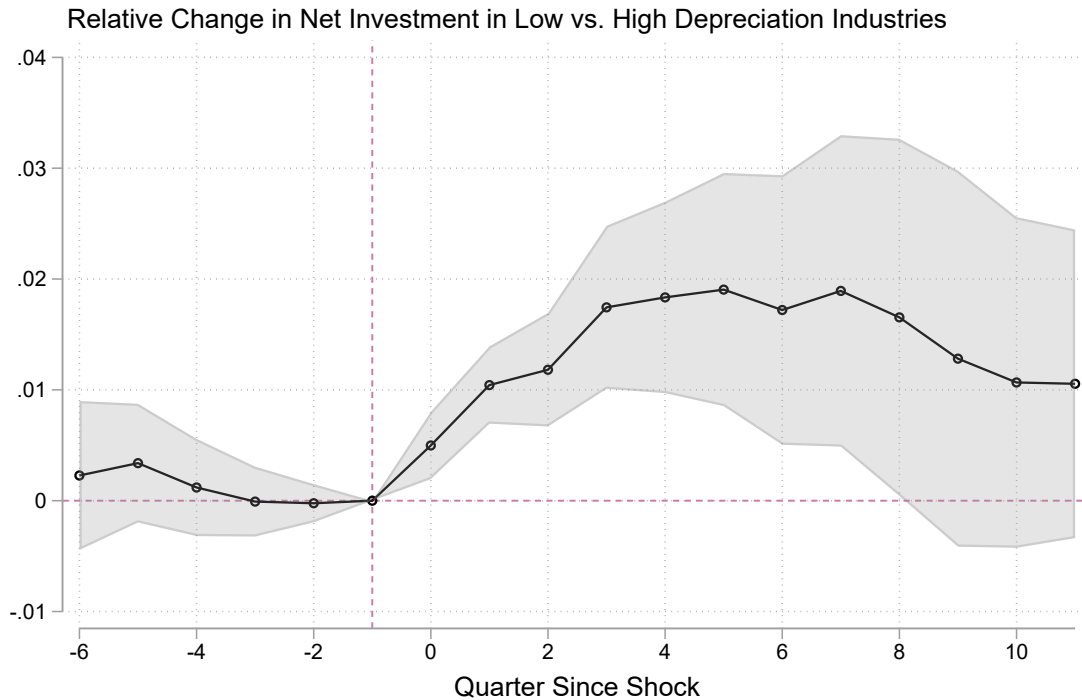
Figure 3 shows that after an expansionary monetary policy shock, firms with lower depreciation rates increase investment more than firms with higher depreciation rates, consistent with the theoretical model introduced in Section 2. Before the monetary shock, firms with different depreciation rates show similar investment trends. The pre-shock coefficients are all very close to zero, statistically insignificant, and show no evidence of pre-existing trends, validating the identification assumptions presented in Section 2. In the quarter of the shock, this flat trend changes sharply: low-depreciation industries start increasing their investment at substantially faster rates than high-depreciation industries. This faster investment growth continues until about one year after the monetary shock, when the coefficients flatten and plateau. The coeffi-

¹⁸The only control for the business cycle in our baseline specification is that we control for interactions of whether the economy is an expansion or contraction, with exposure to both the depreciation rate and the supply of capital goods. However, we show in Appendix Figure A8 that the results are remarkably consistent when not including these controls.

¹⁹However, both alternative shocks have less power, because using the narrative shocks in Romer and Romer (2004) requires us to drop more than a decade of our sample, while the shocks identified through an AR(4) process for the real rate have much less persistent effects on real rates. However, across both alternative shocks, we can reject that interest rate changes have no effect on investment. We can also reject the null provided by the Zwick and Mahon (2017) evidence on user-cost elasticities at all horizons.

cients remain at this level until about two years after the monetary shock, when they begin to decline back towards zero. The coefficients are statistically significant at the 5% level starting in the quarter of the shock and remain significant until about nine quarters after the shock.

Figure 3: Monetary Policy Shocks Increase Net Investment



Notes: This figure plots the β_h coefficients from Equation 6. The coefficients represent the Jordà (2005)-style impulse response for the effect of monetary policy shocks on net investment, comparing low vs. high depreciation firms. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

The magnitude of the effect is economically significant: consider a quarter with a 25 b.p. monetary shock, implying that nominal rates fell immediately by 0.25 p.p. One year after the shock, the estimates in Figure 2 imply that short-term safe interest rates will have fallen by approximately 0.6 percentage points. In response, the capital stock becomes about 0.4% larger in firms with depreciation rates at the sample mean ($\delta_i = 8.4\%$) relative to a firm with a depreciation rate that is one standard deviation above the mean ($\delta_i = 11.7\%$). Table 1 shows that this increase is substantial relative to average capital growth during the sample period. On average, firms increase their capital by 2 log points over one year, implying that the effect size is approximately 18% of the average capital growth rate after a substantial monetary shock.

Overall, I find that monetary policy has significant effects on capital accumulation. This is consistent with a model where a fall in interest rates reduces the cost of capital and leads firms

to increase investment spending. The results suggest that the overall effect of monetary policy on investment, including policy changes priced into financial assets outside of the window immediately surrounding Fed announcements, is likely to be economically substantial.

4.4 Magnitudes and Comparison to Past Work

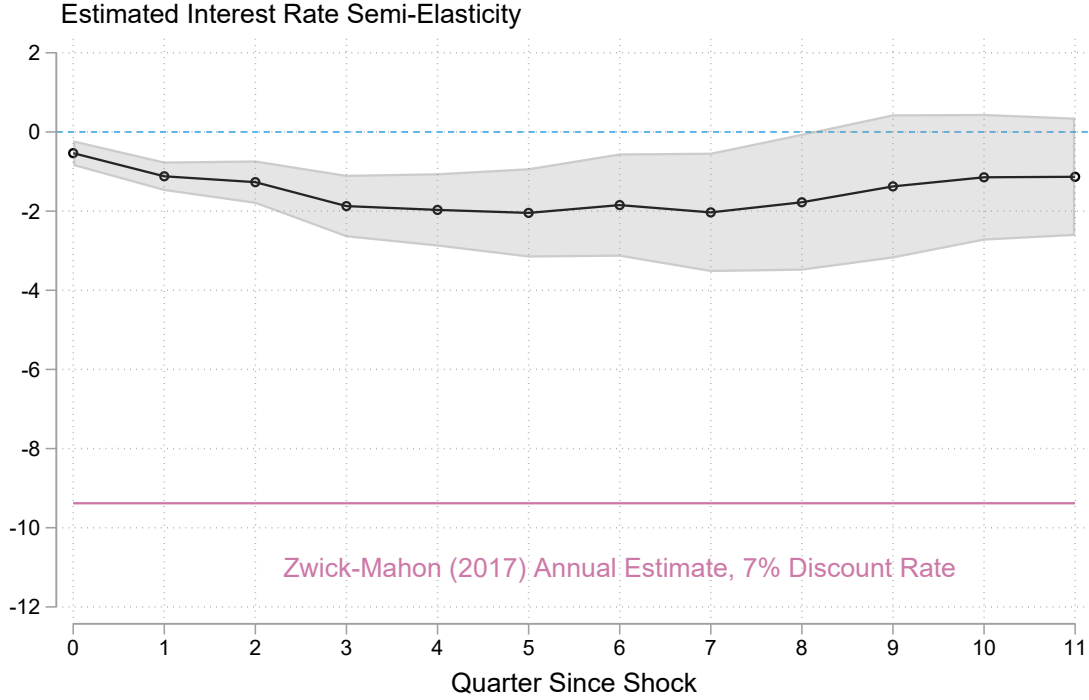
Section 2 derives a series of simple formulas to convert the empirical results into interest rate and user cost elasticities that can be compared to previous work and used to discipline quantitative models. Converting the β_h coefficients into user cost and interest rate elasticities requires data on the components of the user cost of capital: $\mu_t = R_t + \delta + \Theta$. I take the average industry-level depreciation rate based on the merged BEA-Compustat sample (approximately 10% per year) as δ .

In contrast, Θ , the discount-rate wedge, is unobserved. I follow Zwick and Mahon (2017) in assuming that firms discount cash flows at 7% annually ($R_t + \Theta = 7\%$). Figure 4 plots the rescaled coefficients from Figure 3, converted into an interest rate semi-elasticity using Equation 3. I scale the β_h coefficients by the peak log change in the user cost implied by a one-unit monetary shock. Thus, these coefficients represent the percent change in total capital in response to a 1 p.p. increase in real interest rates.²⁰ Figure 4 shows that the estimates in Figure 3 imply significant partial equilibrium interest rate semi-elasticities. One year after the shock, the estimates imply that a one percentage point decrease in real interest rates would increase capital demand by about 2%. The total effect on capital demand remains constant at about a 2% increase for another year before declining slightly. Overall, these estimates provide a new moment that can be used to discipline models of capital investment.

While there are few other estimates of the partial equilibrium interest rate semi-elasticity estimated directly from changes in interest rates, the investment response to temporary changes in tax policy can be informative about the response to interest rates. Consider a version of the firm problem from Section 2, which contains features of the corporate tax code. Hall and Jorgenson (1967) show that, without adjustment costs, the tax-adjusted user cost is then given by:

²⁰This calibration provides similar results for the headline user cost and interest rate elasticities as rescaling by the peak effect from a regression of the monetary shock interacted with the depreciation rate on the log change in the user cost (after residualizing on sector-by-quarter fixed effects and the same controls as the baseline), where I set Θ so that $R_t + \Theta = 7\%$ on average over the sample period. Appendix Figure A4 plots the coefficients from that impulse response. Appendix Figure A4 shows that, six quarters after a one-unit monetary shock, each standard-deviation decrease in the depreciation rate causes the user cost to fall by about 3 log points. At a one-year horizon, this user-cost response yields a user-cost elasticity of -0.5 versus -0.3 in our baseline calibration. Similarly, rescaling by the horizon-specific response generates a one-year interest rate semi-elasticity of about -3 and a user-cost elasticity of about -0.5, because the peak interest rate response occurs at $t = 6$.

Figure 4: Comparing the Estimated Interest Rate Semi-Elasticity to Estimates from Tax Policy



Notes: This figure plots the estimated interest rate semi-elasticities based on the estimates in Figure 3. The graph also features lines indicating the interest rate semi-elasticity derived from Zwick and Mahon (2017).

$$\mu_{f,t} = (R_{t+1} + \delta_f + \Theta) \frac{1 - \tau \zeta}{1 - \tau}$$

Here, τ is the corporate tax rate, and ζ is a parameter that represents the generosity of the tax depreciation system.²¹ If $T = \frac{1 - \tau \zeta}{1 - \tau}$ represents the effect of corporate taxes on the user cost, then the response of capital to changes in taxes is analogous to changes in interest rates:

$$\frac{\partial K_t / K_t}{\partial T / T} = \Psi \quad (7)$$

²¹As noted in Auerbach and Gale (2022), changes in interest rates may also change the user cost of capital by altering the rate at which firms discount tax depreciation benefits. While this may affect the magnitude of Ψ , I abstract from it in calculating Ψ for three reasons. First, Zwick and Mahon (2017) show that bonus depreciation stimulates investment only through immediate deductions, not by changing the PDV of depreciation deductions, implying that the effects on investment from this channel are empirically small. Second, in a calibration that matches the empirical values of the PDV of depreciation deductions from Zwick and Mahon (2017), I find that the “direct effect” of small changes in interest rates on the user cost is approximately 8x greater than the “indirect effect” coming through changes in the PDV of depreciation deductions. Finally, changes in user cost will be picked up only if changes in the PDV of depreciation deductions vary more for long-duration industries than for short-duration industries. In practice, the relationship is unclear ex-ante, and potentially non-linear (Auerbach and Gale, 2022).

Thus, the response of capital investment to changes in tax rates identifies the same parameter as Equation 4 estimated in this paper. This is because in partial equilibrium, both tax rates and monetary policy affect capital investment through their effect on the user cost. I thus compare the estimates from this project to recent estimates from Zwick and Mahon (2017), who estimate the effect of temporary changes in ζ on investment.

The pink line in Figure 4 shows that, while economically significant, the implied semi-elasticities estimated from monetary shocks are much smaller than the corresponding estimates from temporary changes in tax policy. The user cost elasticities estimated from monetary policy in this project are about -0.3 at both the one- and two-year horizons. In contrast, in an annual panel regression, Zwick and Mahon (2017) estimate an elasticity of -1.6, which is roughly five times as large in magnitude. The results in Zwick and Mahon (2017) suggest that increasing interest rates by 1 percentage point would reduce capital demand by around 9.4%, compared to the 2% estimated here. In addition, the estimates from Zwick and Mahon (2017) are not only much larger economically, but the point estimate from Zwick and Mahon (2017) also lies well outside the implied elasticity from the lower bound of the 95% confidence interval in Figure 4. As a result, I estimate that the partial equilibrium effect of interest rates on investment, while economically significant, is substantially smaller than the benchmark Zwick and Mahon (2017) estimate. While the results in Zwick and Mahon (2017) are most directly comparable to my own because both monetary policy and the first rounds of bonus depreciation policy that they study were explicitly designed as temporary policies, my estimate of a user cost elasticity of -0.3 is at the lower bound of the range of estimates surveyed in Chodorow-Reich (2025), which include one-year responses to both permanent and temporary tax policy changes.

My results are also directly comparable to recent estimates from Best, Born and Menkhoff (2026). They estimate a partial equilibrium investment elasticity using a high-quality German survey which directly asks managers how they would adjust investment behavior in response to a change in interest rates. To compare the capital demand elasticity with the investment elasticity reported in that paper, I convert my capital elasticity into an investment elasticity using average cumulated four-quarter investment spending divided by the lagged capital stock. I estimate an investment semi-elasticity of -8 at a one-year horizon, implying that a 1 p.p. increase in interest rates reduces investment by 8%. Best, Born and Menkhoff (2026) estimate a one-year investment semi-elasticity of -6. Thus, my estimates are very similar to those in Best, Born and Menkhoff (2026), with both papers finding robust evidence of lower investment elasticities than those implied by Zwick and Mahon (2017).

The results imply that while interest rates have a meaningful effect on investment, they are substantially smaller than the corresponding effect of temporary changes in tax rates. Since monetary and fiscal policy have very different effects on investment behavior, using estimates

from tax policy to discipline the interest rate elasticity will likely lead to an overestimate of the impact of monetary policy on investment. These results also present a puzzle for traditional neoclassical models, which suggest that the exact same parameter determines tax rate elasticities and interest rate elasticities.

4.5 Which Interest Rates Matter for Investment?

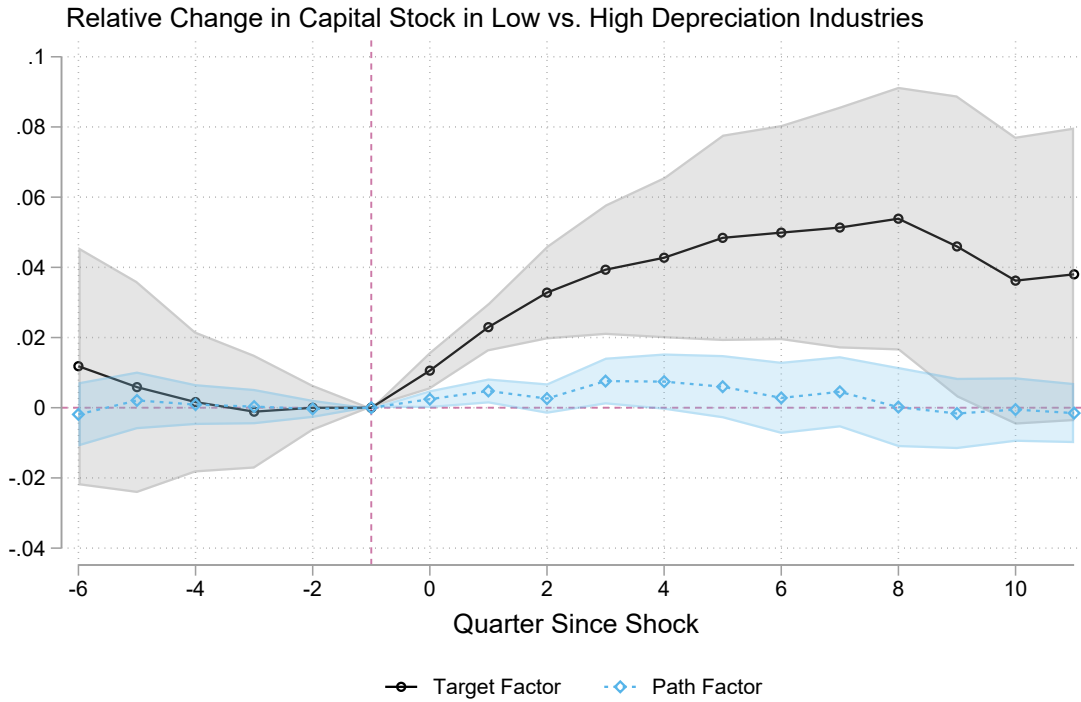
In neoclassical models without adjustment costs, the only interest rate that matters for investment decisions is the short-term, one-period interest rate. In models with more complicated frictions, such as adjustment costs, the whole path of short-term rates affects the magnitude and timing of firm investment (Auerbach and Hassett, 1992). However, in models where the expectation hypothesis does not hold²² or where firms use longer-term debt to finance investment, long-term rates and changes in risk may affect investment choices. To investigate these potential channels, I follow Gürkaynak, Sack and Swanson (2005) and look separately at the investment response to monetary shocks affecting different interest rates. Gürkaynak, Sack and Swanson (2005) separate monetary shocks into two components: a “target” factor that corresponds to surprise changes in the current federal funds rate and a “path” factor that contains information about the future path of policy that is orthogonal to the current target. Gürkaynak, Sack and Swanson (2005) show that the path factor contains information about future rates that strongly affect the long end of the yield curve, including the five- and ten-year Treasury yields.

Figure 5 shows the investment response separately to the target and path factors. Following Gürkaynak, Sack and Swanson (2005), I scale the target factor so that a 1-unit shock to the target factor corresponds to a 1 p.p. decrease in the Fed Funds Rate target. I then scale the path factor to have the same effect on the one-year-ahead eurodollar future as the target factor. I then sum all shocks occurring within a quarter. As a result, the units of the shock have a different magnitude and interpretation than the baseline estimates.

Figure 5 shows that the investment response is driven by shocks to short-term interest rates. The response to the target factor follows the timing of the baseline results, rising sharply in the quarter of the shock, peaking around two years after the shock, and then declining back toward zero. In contrast, the response to the path factor is about one-fifth as large in magnitude and statistically significant at only a few horizons. These results suggest that the investment response is due to unexpected changes in information about the immediate short rate and path of short rates rather than information about longer-term rates.

²²This would include any model where long-term interest rates are not exclusively determined by expectations about short-term rates.

Figure 5: Response of Investment To Target and Path Factors



Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level outcomes. The outcome variable is the change in net investment. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

This decomposition also addresses a potential concern for the identification strategy: that monetary shocks disproportionately lower long-term rates, which are more relevant for longer-duration firms that issue longer-term debt due to maturity matching. If this were the case, we would expect the results for the path factor to be larger than the results for the target factor because the path factor affects longer-term rates that are more relevant for corporate debt financing. In contrast, we find much smaller investment effects from shocks to the path factor, which provides additional confidence that our results are driven by the neoclassical interest-rate channel. Moreover, Appendix Figure A3 shows that I find precise null effects of our measure of exposure on either total debt or long-term debt, suggesting that long-term debt is not the marginal source of financing for the investment responses to these interest-rate shocks. Overall, the results from the path factor, along with the limited response of longer-term debt, suggest that our results are driven primarily by changes in short rates, consistent with the neoclassical interest-rate channel of investment.

4.6 Estimating the Investment Response in the Entire U.S. Economy

One potential concern with comparing the empirical results in this project to those in Zwick and Mahon (2017) is that this project uses the quarterly Compustat data, while Zwick and Mahon (2017) use annual IRS data, which includes a broader sample of firms, including small, private firms. Zwick and Mahon (2017) show evidence that smaller firms have larger investment elasticities. If these firms are underrepresented in Compustat relative to the IRS panel, it could bias my estimates downward relative to Zwick and Mahon (2017). To address this concern, I estimate the effect of both bonus depreciation and interest rate policy simultaneously in a broader dataset covering investment for the entire U.S. economy at the capital-good level. This allows me to compare the effects of tax and monetary policy in the same dataset and time period. I use data on aggregate investment by capital good type from BEA, as well as data from House and Shapiro (2008) and Zwick and Mahon (2017), to create a panel of nominal investment spending at the capital-good level combined with data on rates of economic and tax depreciation.

I estimate the following regression specification:

$$\log(I_{i,t}) = \beta \epsilon_t \delta_i + \alpha_i + \alpha_{c,t} + \eta \frac{1 - \tau z_{i,t}}{1 - \tau} + e_{i,t} \quad (8)$$

In this specification, $i \subseteq c$ indexes a capital good that is part of an investment good class: $c \in \{\text{structure, equipment}\}$. The dependent variable is gross nominal investment spending for capital good i in time t . The primary coefficient of interest is β , which is the coefficient on the interaction of the monetary policy shock ϵ_t , and the capital good's standardized rate of economic depreciation. The specification also includes α_i , a capital good fixed effect, and $\alpha_{c,t}$ is a capital good class-by-time fixed effect. This estimator thus restricts comparisons to goods that are either equipment or structures but face different economic depreciation rates and, thus, differing investment responses to monetary policy. Finally, $\frac{1 - \tau z_{i,t}}{1 - \tau}$ is the tax term that estimates the effect of bonus depreciation policies on nominal investment during this time period. Because relatively few capital goods are covered in this quarterly BEA dataset, I cluster standard errors at the time level only.

Table 3 shows the effect of monetary policy and tax policy on investment in the BEA dataset. The bottom row converts the β coefficient into a user cost elasticity that can be directly compared to the tax term elasticity. Reassuringly, the results show that monetary policy has significant effects on investment. The results are insignificant prior to the shock quarter, after which investment is persistently higher for capital goods with lower rates of economic depreciation compared with higher rates of economic depreciation. The results also show a significant effect

of bonus depreciation.²³

Table 3: The Effect of Bonus Depreciation and Monetary Shocks on Gross Investment

	Gross Investment					Cumulative Investment	
	$t=-4$	$t=-2$	$t=0$	$t=2$	$t=4$	$t=4$	$t=8$
Exposure x Shock	-0.0077 (0.023)	0.029 (0.029)	0.042* (0.025)	0.063*** (0.024)	0.070*** (0.023)	0.056*** (0.019)	0.045** (0.019)
Tax Rate UCE	-4.46*** (0.44)	-4.35*** (0.47)	-4.26*** (0.43)	-4.17*** (0.43)	-4.02*** (0.40)	-4.10*** (0.38)	-3.90*** (0.36)
N	3132	3204	3240	3168	3096	3240	3240
R^2	0.92	0.92	0.92	0.92	0.92	0.93	0.93
Capital FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date-Class FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Interest Rate UCE	0.095	-0.36	-0.52	-0.78	-0.86	-0.69	-0.55

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows the effect of interest rate and bonus depreciation shocks in BEA capital goods data. The first row shows the effect on the main coefficient of interest, the depreciation rate interacted with the monetary policy shock. The second row shows the effect on the tax term user cost elasticity. The bottom row converts the monetary shock interaction term into an interest rate user cost elasticity that can be compared to the tax term user cost elasticity. The first five columns show the effect of the shocks on gross investment (in logs) at different horizons. The last two columns show the effect on cumulated gross investment (in logs) at two different horizons. Standard errors clustered by time in parentheses.

However, across all time periods, the effect of bonus depreciation is substantially larger than that of monetary policy. The tax term elasticities for bonus depreciation are approximately -4, while the corresponding elasticities for monetary policy lie between -0.5 and -1.²⁴ After the shock, across all horizons, we can reject that the bonus and monetary policy elasticities are equal with $p < 0.01$. As with the baseline investment results using Compustat, I find that at around a one-year horizon, the tax rate response is roughly 4-5 times larger than the interest rate response. The BEA results are consistent with interest rate semi-elasticities of investment between 3 and 5, implying that raising interest rates by 1 p.p. lowers investment by 3-5%. These are similar to the baseline results in Best, Born and Menkhoff (2026) and slightly smaller than the baseline results from the Compustat sample. Overall, the results from the BEA data provide additional evidence that monetary policy has significant effects on investment and that tax policy has substantially larger effects than monetary policy.

²³The fact that the pre-period coefficients are significant for the tax term coefficient likely reflects the fact that bonus depreciation policies are autocorrelated over time, unlike monetary shocks.

²⁴Note that these elasticities fall right within the consensus range for the tax user cost elasticity suggested by Hassett and Hubbard (2002).

An alternative way to increase the comparability of the Compustat estimates to those from Zwick and Mahon (2017) would be to reweight the sample to more closely approximate the full U.S. firm-size distribution. Zwick and Mahon (2017) finds a larger response of small firms in both IRS and Compustat data, and thus that reweighting Compustat estimates can mimic the estimates in the full IRS sample. However, and in contrast to the findings from the bonus depreciation literature, I find very limited heterogeneity in the investment response to interest rate policy by firm size. Appendix Table A2 estimates interest rate elasticities after splitting each firm-quarter observation into quintiles based on one year of lagged sales after residualizing on industry and date fixed effects. None of the interaction terms between size quintiles and monetary-policy exposure is statistically significant, and the estimated coefficients do not move monotonically with firm size.

Appendix Table A3 more closely follows the methodology in Zwick and Mahon (2017) and splits the sample into small and large firms based on each firm’s most frequent classification above or below the median of residualized sales over the entire sample. At the one-year horizon, the response to interest rate shocks is nearly identical for large and small firms. At the two-year horizon, the response is larger for large, rather than small firms. Overall, I find no consistent evidence that smaller firms respond more strongly to interest rate changes than large firms, unlike the pattern documented for bonus depreciation. If anything, the results from Appendix Table A3 suggest that reweighting the estimates from monetary shocks to match the sample in Zwick and Mahon (2017) would lead to smaller, not larger, interest rate responses at later horizons. This finding is consistent with the representative BEA results, which show smaller investment responses than the Compustat sample, although the results are qualitatively similar. Thus, reweighting our estimates to more closely match the universe of U.S. firms would, if anything, lead to even wider discrepancies between the tax rate and interest rate elasticities.

Finally, Appendix Table A4 tests whether the larger investment response for low depreciation firms translates into higher prices for capital goods. This would be consistent with upward-sloping capital supply curves functioning as an external adjustment cost offsetting lower interest rates.²⁵ To test this, I reestimate Equation 8, but use the log nominal price of each good as the dependent variable. Appendix Table A4 shows no significant effect of monetary shocks on the prices of capital goods at any horizon. The results suggest that monetary shocks do not increase prices, implying that external adjustment costs are not significant factors affecting the responsiveness of investment.

²⁵External adjustment costs in the form of upward-sloping capital supply curves are common features of macroeconomic models (Ottonello and Winberry, 2020) and underlie other important macroeconomic channels, including the financial accelerator channel (Bernanke, Gertler and Gilchrist, 1999).

Appendix C shows a bevy of additional results that confirm the robustness of the baseline estimates. We estimate a wide variety of specifications, including changing the outcome variable, the sample period, and the set of control variables. Across all specifications, we find similar qualitative patterns: lower interest rates lead to higher investment among low-depreciation firms, with responses quantitatively similar to our baseline estimates. Overall, the robustness tests confirm the baseline finding that interest rates have a significant effect on firm-level investment, but the magnitude of the response is significantly smaller than the response to changes in tax rates.

5 Model

Empirically, this paper provides a new estimate of the interest rate elasticity of investment and shows that it is substantially smaller than the elasticity of investment to changes in corporate taxes. This poses a challenge for standard neoclassical models, where the same parameter governs the response to both types of policies. In this section, I build a quantitative model of capital investment with heterogeneous firms to highlight how the empirical estimates from this paper can be used as moments to discipline quantitative magnitudes and adjudicate between different macroeconomic models. The model builds closely on the heterogeneous firm blocks of Ottonello and Winberry (2020), Fang and Zhou (2025), and Koby and Wolf (2020), but augments these models by including a corporate tax system that allows me to study the effects of both monetary and tax policy.

The model is a dynamic extension of the model presented in Section 2, but includes a second factor of production, a corporate tax code, and more flexible financial frictions and adjustment costs. A fixed mass of firms indexed by the unit interval produces a homogeneous good using a decreasing returns to scale production technology. Firms vary both in their idiosyncratic productivity and in the rate of capital depreciation. Firms have idiosyncratic productivity draws (z), which follow an AR(1) process. In addition, firms draw a constant depreciation rate (δ), which is taken as given. Thus, the model is designed to match the assumption underlying the empirical strategy: firms are endowed with a production process based on their industry, which determines the nature of the capital they use in production.

Firms face an infinite horizon and solve a capital accumulation problem. Because I find empirically that firms do not use debt financing to increase investment spending after monetary shocks, I do not allow firms to use debt to fund investment. However, in the baseline model, firms can pay out negative dividends to fund higher investment (equity issuance), though this can be subject to a cost in the presence of financial frictions. The firm's state variables are its productivity, capital stock, depreciation rate, as well as their stock of tax-depreciation deduc-

tions, which I use to model the corporate tax code. In each period, the firm takes its productivity as given, chooses next period's capital, hires labor in a spot market, and then produces. Firms maximize a stream of (net-of-tax) dividends discounted by the interest rate. Formally, the firm problem is:

$$V_t(k, z, B, \delta) = \max_{k', n} p_t f(k, z, n_t) - w_t n - q_t * (k' - (1 - \delta) * k) - AC(k', k, I_t; \zeta) - T(\pi_t, \tau_t, b_t, B, I_t) - \Phi(d_t^0, \pi_t, p_t, Y_t) + \mathbb{E}_t \left[\frac{1}{1 + r_t} V_{t+1}(k', z', B', \delta) \right] \quad (9)$$

where $Y_t = f(k, z, n_t) = e^z k^\alpha n_t^\nu$, $\pi_t = p_t Y_t - w_t n_t$, $d_t^0 = \pi_t - q_t I_t - AC(k', k, I_t; \zeta) - T(\pi_t, \tau_t, b_t, B, I_t)$, and $I_t = k' - (1 - \delta) * k$.

In this model, $AC(k', k, I_t; \zeta)$ is a capital adjustment cost that can include both fixed and convex costs, $\Phi(d_t^0, \pi_t, p_t, Y_t)$ are the costs associated with financial frictions, $\frac{1}{1+r_t}$ is the firm's discount factor, q_t is the price of capital, and p_t is the price of the goods produced by the heterogeneous firm.

T is a function that captures the flow of corporate taxes owed by the firm in each period. The firm's corporate tax bill is a function of two potentially time-varying parameters: the corporate tax rate τ_t and a parameter b_t , which captures the generosity of bonus depreciation. Modeling a realistic tax depreciation system requires keeping track of the tax benefits owed to firms by the government in future periods, which are a function of investment choices made in past years. Traditionally, researchers have used the fact that since previous investment decisions are sunk, they should not influence current investment choices that firms make beyond their effect on the current capital stock (Winberry, 2021; Chen et al., 2023). However, in the presence of external-financing constraints, this is no longer true because the timing of payments affects whether and how quickly firms hit their financial constraint. Thus, I model the tax depreciation system using a single state variable that represents the stock of deductions the firm has going into each period. Total taxes depend on profits net of deductions: $T(\pi_t, \tau_t, b_t, B, I_t) = \tau_t(\pi_t - D_t)$, while the firm's tax deduction in the current period is given by: $D_t = \delta_{\text{tax}} B + b_t q_t I_t$. Thus, bonus depreciation policies are valuable because they allow the firm to immediately deduct a fraction b_t of its investment costs. Finally, the stock of deductions evolves according to: $B' = B + q_t I_t - D_t$.²⁶

The empirical estimator in Section 4 includes sector-by-time fixed effects, which difference out general equilibrium factors that affect all firms. Thus, I initially solve the model in partial equilibrium and match the empirical response of investment to shocks estimated in this paper and Zwick and Mahon (2017). Appendix D shows how to insert the heterogeneous firm problem in a full, general equilibrium New Keynesian model.

²⁶As a simplification, I make taxes fully refundable to avoid having to track additional state variables representing firm's history of losses.

Taking initial prices $\{p_t, w_t, q_t\}$, policy parameters $\{r_t, \tau_t, b_t\}$, and an initial firm-size distribution $\mu_0(k, z, B, \delta)$ as given, an equilibrium of the model is a set of value functions $V_t(k, z, B, \delta)$, policy functions $k_t(z, k, B, \delta)$, and a firm-size distribution $\mu(k, z, B, \delta)$, such that all firms optimize and the distribution of firms is consistent with the decision rules. I then introduce temporary shocks to fiscal and monetary policy parameters in the model to study investment dynamics in response to monetary and fiscal policy shocks, while holding other prices ($\{p_t, w_t, q_t\}$) fixed. I study the response of investment and other firm-level variables to these shocks and the transition back to steady state.

To study monetary policy, I solve for the transition path after a series of shocks to the real interest rate r_t that matches the path of real interest rates estimated in Figure 2. To study how investment responds to changes in corporate taxes, I study the response to a shock that mimics the first round of bonus depreciation. I solve for the transition path after a policy change realized at time zero, where the expensing parameter b_t changes from zero to 0.30 and lasts for 13 quarters.²⁷ For both shocks, I assume all agents have exact knowledge of the sequence of shocks when the first shock is realized. As a result, I study a perfect-foresight transition path back to the steady state and solve the model using the methodology described in Boppart, Krusell and Mitman (2018). As a result, the transition path in my model is fully nonlinear.

6 Quantitative Results

In this section, I bring the empirical estimates of the investment response to interest rates from Section 4 to the model. I consider two distinct types of models with frictions that can be calibrated to match the empirical impulse response of investment. First, I consider a model with adjustment costs, calibrating the strength of the adjustment costs to match the investment response. Then, I consider a model without any adjustment costs, but with financial frictions. In practice, this limits the amount of equity the firm can issue each period.

6.1 Calibration

Table 4 shows the externally calibrated parameters that remain consistent across all calibrations of the model. The model is at the quarterly frequency. I set the discount factor $\beta = 0.99$. The coefficients on capital (α) and labor (ν) in the production function are equal to 0.28 and 0.57, respectively. This implies a 33% capital share and a decreasing returns-to-scale parameter of

²⁷However, I allow the shock to taper off and return gradually over four quarters, rather than immediately returning to zero at the end of the policy. This avoids very sharp investment spikes in the frictionless version of the model.

approximately 0.85. I set the persistence and dispersion of the idiosyncratic TFP process to match the estimates in Winberry (2021).

The baseline model has two possible depreciation rates equal to 1.7% or 3% per quarter. Finally, I calibrate the baseline corporate tax rate and the baseline expensing parameter to match the U.S. corporate tax code before the first round of bonus depreciation. I also set the baseline tax depreciation rate to match an approximate 7-year recovery period under the MACRS tax depreciation system.

Table 4: Externally Calibrated Parameters

Parameter	Description	Value	Source
β	Discount Factor	0.990	Data
ν	Labor Coefficient	0.57	Labor Share
α	Capital Coefficient	0.28	Capital Share
ρ	Persistence of TFP	0.90	Winberry (2021)
σ_j	S.D. of TFP Innovations	0.053	Winberry (2021)
τ	Baseline Corp. Tax Rate	0.35	Pre-TCJA Statutory Rate
δ_{tax}	Quarterly Tax Depreciation Rate	0.0807	7-Year MACRS
$\bar{b}$	Baseline Immediate Expensing Share	0.000	Baseline policy

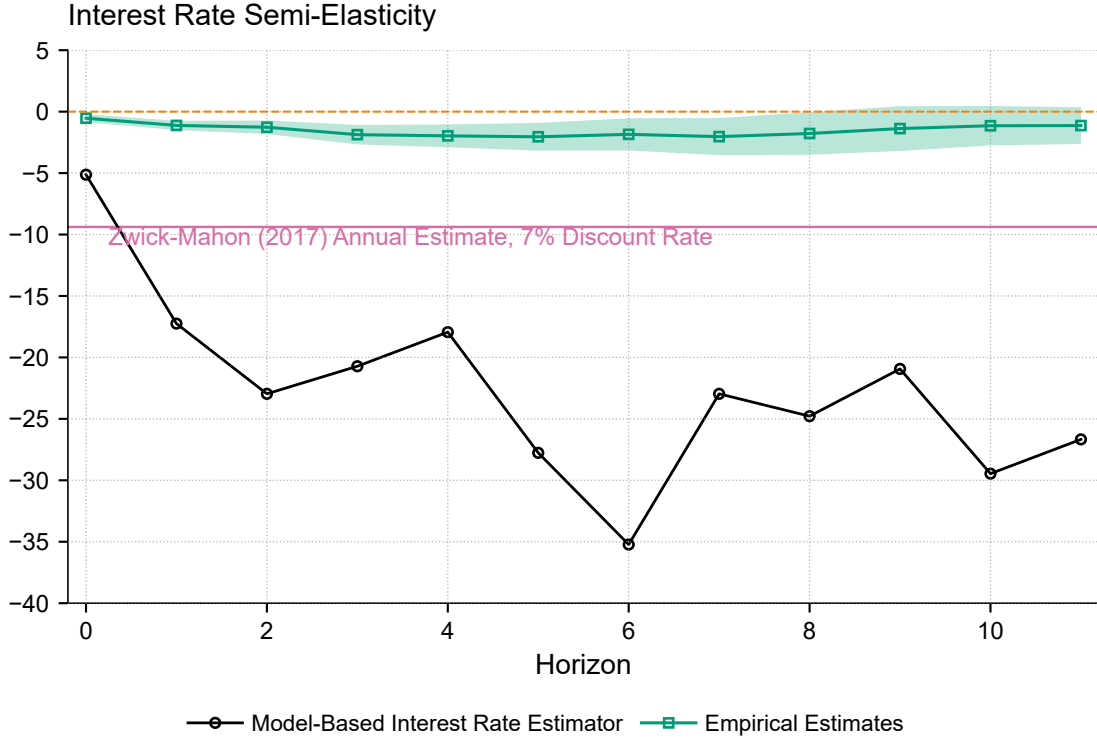
6.2 The Effect of Interest Rate Shocks in a Quantitative Model

First, I investigate how investment responds to interest rate shocks in the quantitative model. I mimic the estimate from the data in the model by taking the total investment response to a monetary policy shock across firms with different depreciation rates. I then rescale the investment response using Equations 3 and 4, just as I do in the data.

First, Figure 6 shows that frictionless models, which abstract from adjustment costs or financial frictions, generate investment responses that substantially exceed the empirical estimates. The black line plots the interest rate semi-elasticities from the frictionless model: a 1 p.p. decline in the real interest rate increases the capital stock by around 20% after one year. Depending on the exact horizon, the estimate is 2 - 4 times as large as that implied by the baseline parameterization from Zwick and Mahon (2017) and roughly an order of magnitude larger than the estimates from Section 4.

These findings highlight the value of partial-equilibrium estimates for constraining the quantitative predictions of economic models. Winberry (2021) and Koby and Wolf (2020) argue that the estimates from Zwick and Mahon (2017) can be used to discipline the magnitude of the investment response to interest rate changes. Although both the estimates from Zwick and Mahon (2017) and my results show that frictionless models imply excessively large

Figure 6: The Effect of an Interest Rate Shock in a Frictionless Model



Notes: This figure reports the response of investment to a monetary policy shock in a version of the model without frictions. The black line shows the aggregate semi-elasticity in the model. The green line and shading show the empirical estimate and 95% confidence interval. The pink line shows the estimate from Zwick and Mahon (2017).

investment elasticities, the discrepancy between them suggests that inferring responses to interest rates from responses to tax policy can substantially overstate the investment response. My estimates thus provide a more direct empirical benchmark for quantitative models of investment responses to interest rates.

Next, I calibrate two separate versions of the model to the empirical interest-rate elasticities from this paper. I first consider the model with adjustment costs. I include both fixed and convex adjustment costs, which have both been argued to be crucial in determining investment dynamics. I adopt the following specification for the adjustment costs.

$$AC(k', k, I_t; \zeta) = \frac{\eta}{2} \left(\frac{k' - k}{k} \right)^2 \frac{k}{\bar{k}_\delta} + \zeta w_t \bar{k}_\delta \mathbf{1} \{ |k' - k| > 0 \}.$$

In this formulation, the parameter η determines the strength of the convex adjustment costs and ζ is a uniform random variable that parameterizes the strength of the fixed cost. I parameterize the adjustment costs in terms of net, rather than gross, investment, implying that firms can do replacement investment without triggering any adjustment cost. The one

nonstandard choice is to normalize both adjustment costs relative to the average steady-state capital stock within each firm's industry. This ensures that adjustment costs in the steady state are similar across industries, even if firms in one industry have larger capital stocks. This is important for matching the data because the model generates very different steady-state capital levels across industries due to the persistent differences in depreciation rates. Though I find that capital stocks are slightly larger in low-depreciation industries, Appendix Figure A1 shows that the data strongly reject the heterogeneity implied by the model. In addition, if adjustment costs varied strongly across industries, this would imply very different dispersions of firm size across industries, as firms in the higher adjustment cost industry would respond less to idiosyncratic productivity shocks. This implication is strongly rejected by the data, as shown in Appendix Figure A1. As a result, higher adjustment costs in low-depreciation industries are not a plausible mechanism for explaining the results. I calibrate both the η and ζ parameters to match the partial-equilibrium semi-elasticity of investment shown in Figure 4.

Next, I consider a variant of the model without adjustment costs but with financial frictions. I model financial frictions as triggering a cost when the firms' dividends fall below a target. Thus, the model approximates a cost for raising equity (i.e., issuing negative dividends), although I also allow the cost to bind before dividends become negative. This reflects that firms may have positive dividend targets, with investors reacting negatively to firms reducing dividends below target, even if total dividends remain positive. Under this model, a firm's cash flows are a primary determinant of the amount of investment they can undertake because (1) firms can use their free cash flows to invest once without any cost when they have hit their dividend target and (2) higher cash flows can directly relax the firm's financing cost. The financing cost takes the following form:

$$\Phi(d_t^0, \pi_t, p_t, Y_t) = \frac{\lambda}{2} \left[\max \left\{ \omega_\delta - \frac{d_t^0}{\pi_t}, 0 \right\} \right]^2 \frac{p_t Y_t}{p Y_\delta}, \quad \omega_\delta = \alpha_\omega \frac{\bar{d}_\delta^0}{\bar{\pi}_\delta}.$$

Here, the parameter λ determines the strength of the financing cost, while α_ω determines the firm's dividend target, below which additional cuts to the dividend (or additional equity financing) incur a convex cost. To make the results more comparable to the adjustment costs model, I have the financing cost scale with firm size, though normalized across industries to again ensure financing frictions across industries are similar in steady state.²⁸ As with the adjustment costs model, I calibrate both the parameters that govern the strength of the financing frictions to match the partial-equilibrium semi-elasticity of investment shown in Figure 4. In the financial-frictions model, I also set the adjustment cost parameters to zero to isolate the

²⁸As with the adjustment costs model, varying financial frictions across industries would imply variance in firm-size distributions across industries that are strongly rejected by the data.

role of financial frictions and distinguish them from real adjustment costs.

Figure 7 shows that both models fit the quantitative magnitudes in the data quite closely. Across both models, the impulse response closely tracks the data until after the peak, when both models are slightly more persistent than in the data, reflecting persistence of the estimated monetary shocks. However, both models require quite strong frictions to match these empirical responses. Naturally, these are necessary because the estimated elasticities are an order of magnitude smaller than those generated by the frictionless version of the model. This highlights how the empirical estimates from this project can discipline models with capital investment and ensure they generate realistic quantitative magnitudes.²⁹ These results also highlight how financial frictions alone can lead to realistic investment elasticities without relying on arbitrary adjustment costs. However, both models can match the magnitudes, suggesting that the investment response to one shock alone is not enough to determine which mechanism better describes the frictions that firms face.

Table 5: Adjustment Costs and Financial Frictions Parameters

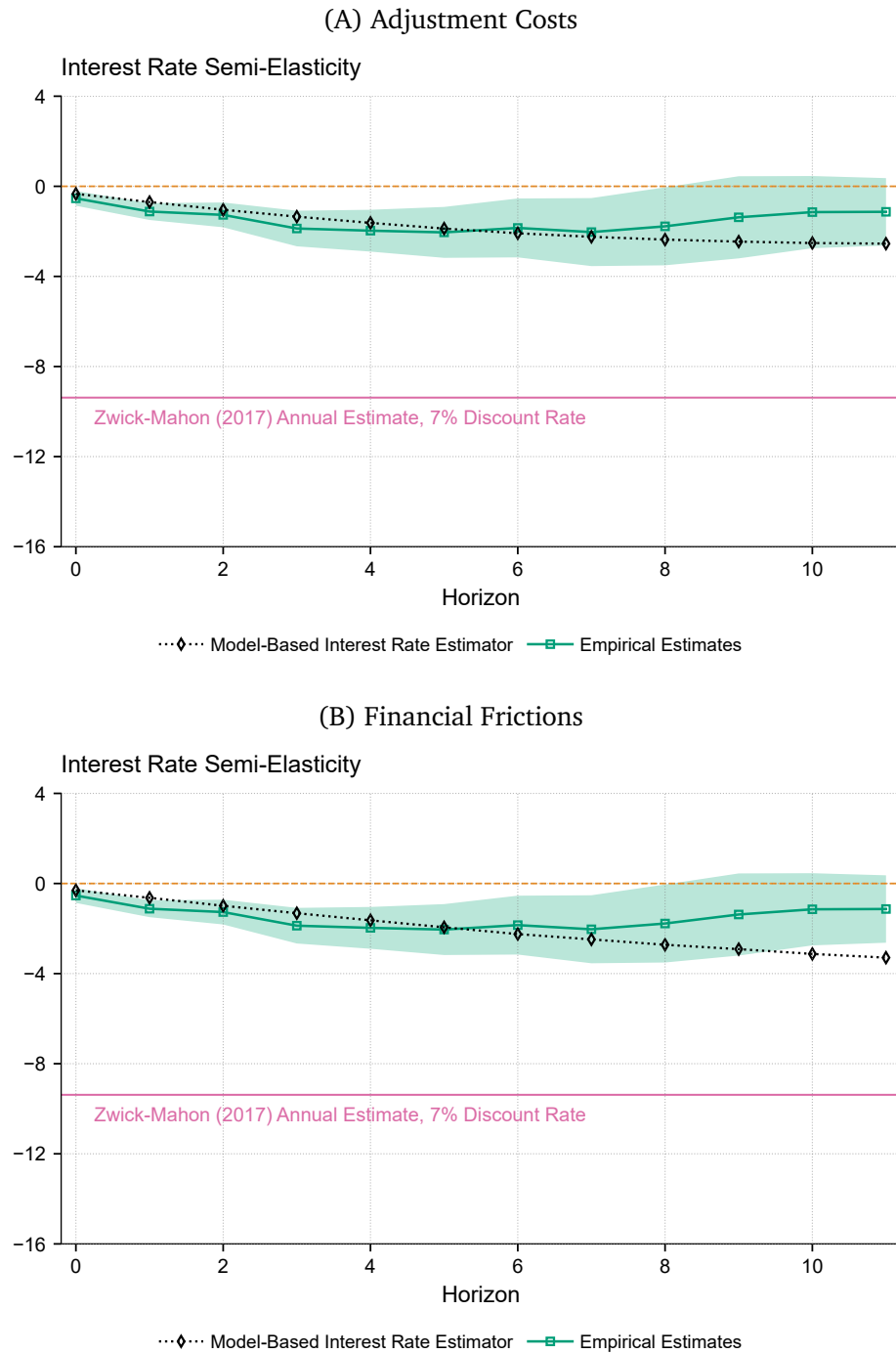
Parameter	Description	Value
Panel (A): Adjustment Costs Model		
η	Convex Cost Parameter	0.018
$\bar{\zeta}$	Fixed Cost Upper Bound	0.40
Panel (B): Financial Frictions Model		
λ	Equity Issuance Cost	0.22
α_ω	Dividend Payout Target	1.0

6.3 The Response of Investment to Tax Rate Shocks

Next, I study the response of investment to tax rate shocks in both models calibrated to match the interest rate elasticity. The response of investment to a tax rate shock functions as an over-identifying test. Since both models can match the magnitude of the interest rate shock, I use the investment response to tax rate changes as an untargeted moment to discriminate between the two models of firm behavior. I study a 13-quarter shock to the expensing parameter, setting $b = 0.30$ instead of $b = 0$ to match the countercyclical, temporary bonus depreciation policies implemented during the 2001 recession.

²⁹Appendix Figure A5 also compares the estimated interest rate elasticities, based on the differences-in-differences estimator, in the model with the actual aggregate investment response. Across both calibrated models, the estimator remains very informative about the actual investment response, confirming the underlying usefulness of the empirical estimator even for disciplining models with very strong frictions.

Figure 7: The Effect of an Interest Rate Shock in the Estimated Models



Notes: This figure shows the estimated interest rate semi-elasticities in both the data and the two versions of the model. The line in green shows the empirical estimates from the data, while the line in black shows the interest-rate semi-elasticity derived from the calibrated model. The line in pink shows the implied elasticity from Zwick and Mahon (2017). Panel (A) shows the results from the model with real adjustment costs, while Panel (B) shows the results from the model with financial frictions.

Table 6: Untargeted Moment: Implied Interest Rate Elasticity from Tax Rate Shock

	Data	Adjustment Costs Model	Financial Frictions Model
One-Year Semi-Elasticity	-9.38	-2.00	-4.67

Table 6 shows the implied interest rate elasticities from the bonus depreciation shock in the two models and the implied estimates from Zwick and Mahon (2017). Table 6 shows that only the financial frictions model generates a tax rate elasticity that is substantially larger than the interest rate elasticity. The adjustment costs model generates a low semi-elasticity of around -2, nearly identical to the interest rate elasticity from a monetary shock but far from the actual tax rate elasticity, which should be larger.

In contrast, the model with financing costs generates elasticities that close a significant share of the gap between the interest rate and tax rate elasticities. The tax rate response in the financial-frictions model implies an interest rate elasticity of -4.7, which is more than double the interest rate elasticity in both the data and the model, and closer to the value of -9.4 implied by Zwick and Mahon (2017). Overall, these results provide support for models in which financial frictions generate smaller investment elasticities and against models that use strong, real adjustment costs to slow the investment response. This also shows how evidence from both tax and interest-rate shocks can be used to discriminate between different models. While calibrations of either model can directly match the estimates from one type of shock, only the financial frictions model can generate heterogeneity in the interest rate and tax rate elasticities highlighted by the empirical estimates in this paper.

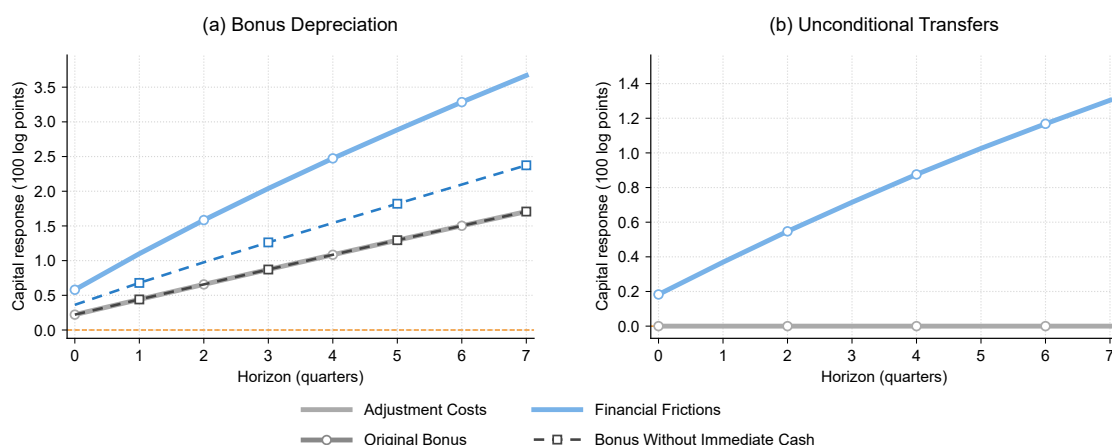
6.4 Underlying Mechanisms

Why doesn't the model with adjustment costs generate substantial discrepancies between tax- and interest-rate elasticities? Recall that in the frictionless, static benchmark, tax and monetary policy operate through the exact same channel in partial equilibrium: both affect firms' investment decisions by changing firms' user cost of capital. This is why the investment response to both policy shocks is governed by the same parameters, and why they should generate identical elasticities. Models with adjustment costs do not change this underlying logic. This is because adjustment costs slow the investment response with the exact same force regardless of which policy instrument is used to change the cost of capital. Thus, adjustment costs alone are not able to generate large differences between the monetary and tax elasticities. This is confirmed in the quantitative model, where the tax and monetary elasticities are quite similar.

In contrast, in the presence of financial frictions, interest rates and tax rates do not operate through the exact same channels. In particular, while both tax rates and interest rates still

have an effect on the cost of capital, in the presence of financial frictions, tax policy has a second effect on investment choices. Lower tax payments also generate extra cash flow for the firm by reducing its immediate tax bill. In a frictionless model, this cash-flow effect is irrelevant because the firm can raise external finance costlessly to pay for its desired investment. However, if borrowing is costly or undesirable, this extra cash flow can be very valuable because the firm can use it to fund its higher desired investment. Because tax policy has this additional channel, the model with financial frictions can generate tax elasticities that are significantly larger than the monetary elasticities, consistent with the empirical estimates.

Figure 8: Mechanism Tests in the Quantitative Model



Notes: This figure reports the response of investment to different policy experiments in the two models. Panel (A) shows the effect of the bonus depreciation shock. The lines in gray show the effects in the adjustment costs model, while the lines in blue show the effect in the financial frictions model. The solid lines show the effect in the original bonus depreciation experiment, while the dashed lines show the effect of the experiment that provides the same tax benefit in present-value terms, without accelerating immediate tax payments. Panel (B) shows the effect of distributing cash lump-sum to firms in each period that are equal in size to the tax benefits of bonus depreciation. In panel (B), the lines in gray show the effect in the adjustment costs model, while the lines in blue show the effect in the financial frictions model.

We use our quantitative model to provide a number of pieces of evidence supporting this interpretation. First, we run an experiment in the model to determine how much of the response to bonus depreciation can be credited to the immediate cash injection that results from bonus depreciation versus the change in the tax component of the user cost. In this experiment, I increase the amount that a dollar of new investment adds to a firm's tax basis until the present value equals the benefit of the immediate and future tax savings under bonus depreciation. This thus provides the same change in the tax component of the user cost as a bonus depreciation policy, without accelerating the immediate cash benefit.

Figure 8, Panel (A) shows the results of this experiment in both the adjustment costs and fi-

nancial frictions calibrations. Notably, in the model with adjustment costs, both the traditional bonus depreciation policy and the policy with deferred tax deductions generate the exact same investment response, confirming that the experiment causes the exact same change in the tax wedge as the full bonus depreciation policy. However, in the model with financial frictions, the version of the tax-policy experiment that does not provide immediate cash benefits reduces the investment response to bonus depreciation after one year by about 38%. In addition, this reduces the difference between the investment response to bonus depreciation in the financial frictions model and the adjustment costs model by about two-thirds, suggesting that the differences across the two models are driven primarily by whether or not firms respond to the additional, accelerated cash flow generated by the tax-policy experiment.

Second, we run an experiment in the model in which we take the tax savings induced by the bonus depreciation policy in each quarter and distribute it to firms as a lump sum. This relaxes the firm's financing friction without affecting the relative price of capital by changing the tax wedge on investment. The results of this experiment are shown in Figure 8, Panel (B). Consistent with the neoclassical model, this has no effect on firms' investment in the model with adjustment costs. However, in the financial frictions model, this causes a large increase in investment, which is again equivalent to about 35% of the full response under the bonus depreciation policy. As a result, both experiments suggest that the significant differences in the response to bonus depreciation under models with adjustment costs and models with financial frictions result from the immediate benefits of additional cash generating an investment response under financial frictions and not adjustment costs. The results of both experiments also show that between 30 and 40% of the investment response to bonus depreciation is generated by the immediate cash rebated to firms, while the rest is due to the traditional changes in the tax component of the user cost.

Figure 8, Panel (B) also provides new evidence on how lump-sum cash transfers affect firm investment. In the quantitative model, bonus depreciation brings forward tax deductions, which also implies that while tax payments are lower in earlier periods, they are higher in later periods. In the model, forward-looking firms respond to both the reduced taxes today and the higher taxes in the future, even though the taxes are lump-sum and thus are invariant to firm decisions. As a result, I consider an experiment where I reduce taxes by the amount of the tax savings when they are positive in the aggregate, but then set the additional transfer to zero in later periods.

This experiment generates a one-year marginal propensity to invest (MPI) of 0.61, implying that for every dollar of tax savings, firms increase investment by 61 cents.³⁰ As noted in Winberry et al. (2025), the MPI is a key parameter for assessing the quantitative relevance of

³⁰When firms forecast higher lump-sum taxes in the future, that leads to even larger responses of 0.74.

New Keynesian models with firm heterogeneity, but we have very little evidence on their magnitude. Thus, my paper contributes new empirical evidence that can discipline the strength of both the interest rate elasticity of investment and the MPI. My estimate of 0.61 is close to the calibrated estimate of 0.50 in Winberry et al. (2025), but is slightly larger than Melcangi (2026)’s estimate of the marginal propensity to hire of 0.39.

Because there was no significant debt response, the model abstracted from firms’ debt choices. In the presence of debt financing, monetary policy can also generate free cash flow by reducing interest expenses. However, the benefits are smaller and less immediate than those for tax financing. Consider a firm that purchases \$1000 of computers, financed with floating-rate debt. If interest rates are reduced by 1 p.p., this reduces the firm’s interest payments by 1 p.p., or around \$10 per year. In contrast, moving to 50% bonus depreciation increases the firm’s free cash flows by \$140 in the year of purchase relative to the no bonus depreciation benchmark, an order of magnitude larger than the effect of monetary policy (Zwick and Mahon, 2017). In addition, because 50% bonus depreciation has a smaller effect on the cost of capital than the 1 p.p. drop in interest rates, and because the majority of corporate debt is fixed rather than floating rate, this example, if anything, understates the differences between tax and monetary policy at generating immediate cash flows. For example, Bräuning, Joaquim and Stein (2023) estimate that a 1 p.p. increase in interest rates only increases interest expense ratios by around 0.5 p.p., with the peak effect occurring more than 1 year after the change in the Fed. Funds rate.

The empirical estimates and quantitative model provide new evidence in support of models with financial frictions where firms’ cash flows are a key determinant of investment financing. This adds new quasi-experimental evidence bolstering a significant literature in economics that has posited an important role for cash flows and retained earnings on investment (Fazzari, Hubbard and Petersen, 1988). This result validates the finding in Zwick and Mahon (2017) that the large elasticities they estimate are driven by taxable firms that receive immediate cash flows as a result of bonus depreciation. In the model, we calculate an MPI of 0.61, which is an important moment in macroeconomic models with firm heterogeneity (Winberry et al., 2025). This also aligns with the evidence in Appendix Figure A3 on financial responses to monetary shocks: firms do not respond to interest rate changes by issuing debt but instead increase cash holdings before increasing investment. This mechanism is also consistent with financial frictions endogenously generating higher firm-level discount rates (Gormsen and Huber, 2025). Gormsen and Huber (2025) quote a CFO who directly ties investment responses and higher discount rates to a cash-flow-based investment constraint that exactly mimics the constraint in the financial frictions model, stating:

“We are living within our cash flow, meaning that we want to be able to fund our

CapEx and our dividend from our cash flow. And so that is the constraint, and so, because we have a limited amount of capital, that is why we have the hurdle rate set at 15 percent IRR for projects."

7 Heterogeneous Responses

The empirical results, when analyzed through the quantitative model, suggest that financial frictions help slow the investment response to interest rate shocks. In this section, I examine heterogeneity in the interest rate elasticity to test this channel more explicitly in the data. I test whether firms that are more likely to be borrowing constrained exhibit smaller interest rate elasticities than those that are less likely to be constrained, consistent with the findings from the quantitative model. In particular, I focus on a specific type of cash-flow-based constraint: interest-coverage covenants, which are the most common debt covenant (Greenwald, 2019). Note that this constraint does not precisely map to the financial friction in the model: that friction placed limits on firms' ability to issue equity, while interest-coverage ratios limit firms' debt issuance. However, both limit a firm's ability to borrow and fund investment, making cash flows inside the firm particularly valuable.

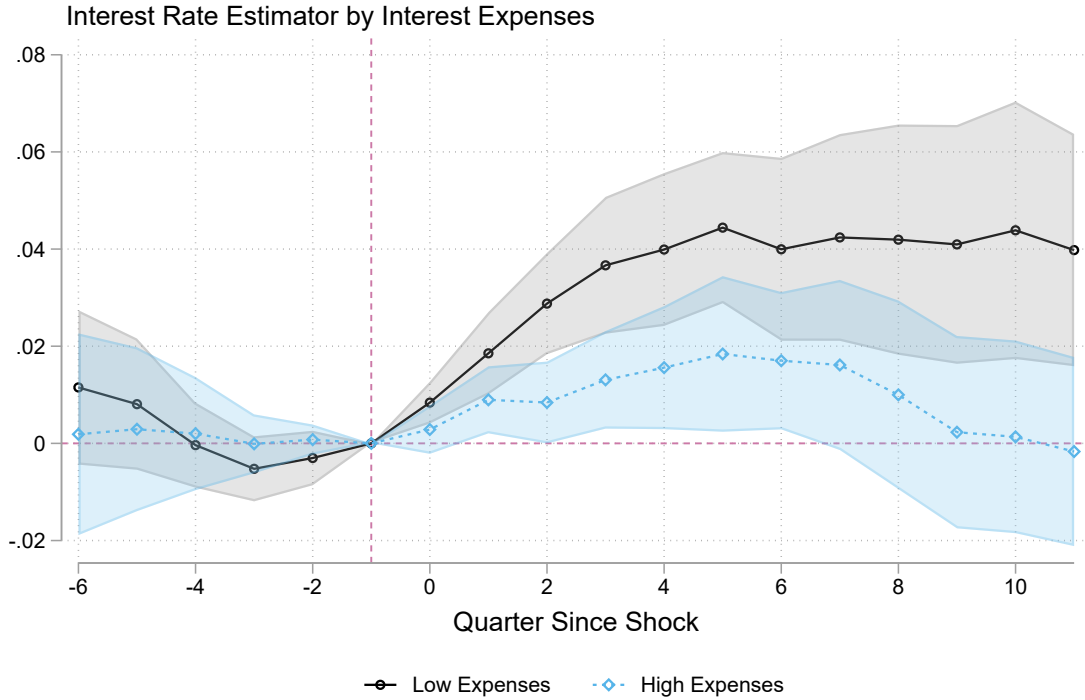
I use a firm's interest expenses as a proxy for whether or not firms are likely to be bound by this financial constraint. In particular, I regress firms' interest expenses on industry and date fixed effects, as well as on splines of total firm assets. I then take residuals from that regression to split the sample into firms with high or low interest expenses.³¹ I then re-estimate the baseline regression specification, allowing for an interaction between the primary variable of interest: the depreciation rate interacted with the monetary shock, and an indicator for whether or not a firm has high vs. low interest expenses.

Figure 9 documents striking heterogeneity by firms' interest expenses. For firms with high interest expenses relative to their assets and industry, the estimated impulse response is close to the baseline. However, for firms with low interest expenses, the effect is significantly larger. The coefficients for firms with low interest expenses are more than twice as large as the baseline and more persistent, implying higher, long-run interest-rate elasticities.

Overall, these results suggest that interest expenses, and thus interest-coverage ratios, are an important determinant of firms' investment responses. This implies that tighter financial constraints are the key constraint limiting firms' investment response to interest rate shocks, consistent with the quantitative model. The results contrast with those from Cao et al. (2025), who estimated limited heterogeneity by interest expenses in Norway using a different empirical strategy, but are more consistent with Modi (2024), who argues that loan covenants

³¹I also remove the middle 20% of firms.

Figure 9: The Effect of Monetary Policy Shocks for Low and High Interest Expense Firms



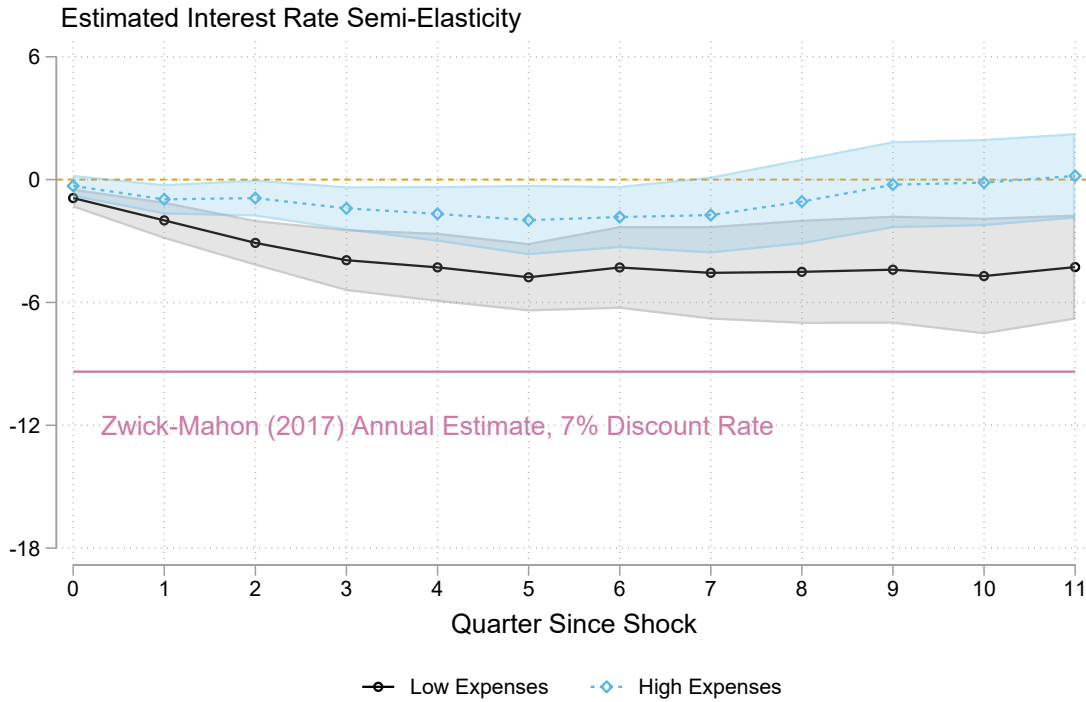
Notes: This figure plots the β_h coefficients from Equation 6, separately for firms with low vs. high interest expenses. The coefficients represent the Jordà (2005)-style impulse response for the effect of monetary policy shocks on net investment, comparing low vs. high depreciation firms. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

play an important role in the transmission of monetary policy through investment. Finally, this strong pattern of heterogeneity provides evidence that the muted baseline results are not driven by sticky discount rates (Fukui, Gormsen and Huber, 2024) or firm inattention to monetary shocks.

Figure 10 converts these impulse responses into semi-elasticities. For low-interest-expense firms that are more likely to be unconstrained, interest rate semi-elasticities are approximately -5 at a long-run horizon, which is much closer to the value of -9.4 estimated in Zwick and Mahon (2017). The magnitude of the heterogeneity is also broadly consistent with the difference between the tax and interest-rate elasticities in the model, providing additional evidence that financial frictions can close a meaningful fraction of the gap between the two responses, though a substantial share remains unexplained. This further supports the mechanism implied by the quantitative model: financial frictions that limit firms' ability to use external funds for new investment lead to smaller interest rate elasticities.

I also examine other frictions hypothesized to affect firms' responses to interest rate shocks.

Figure 10: Interest Rate Elasticities for Low and High Interest Expense Firms



Notes: This figure plots the estimated interest rate semi-elasticities based on the estimates in Figure 9. The graph also features lines indicating the interest rate semi-elasticity derived from Zwick and Mahon (2017).

In particular, I focus on firms' existing leverage (Ottonello and Winberry, 2020), liquidity (Jeenas, 2026), and fixed costs. I follow the same methodology used to study interest expenses for leverage and liquidity. For fixed costs, I split firms by whether or not they had an investment spike in the year prior to the monetary shock.

Table 7 shows results for the heterogeneity tests for these three factors. I only show estimated coefficients one year out from the shock. Column (1) shows the baseline estimates. Column (2) shows the heterogeneous effects of firm leverage. I find the coefficient on the main term is nearly unchanged from the baseline, and the interaction effect is close to zero and statistically insignificant. This provides more evidence in favor of a cash-flow-based constraint against a more traditional financial friction based on firm leverage. It also supports the finding that the split by interest expenses is not driven by firms' proximity to default.

Column (3) also estimates a statistically insignificant effect of firm liquidity on the interest rate elasticity. Though statistically insignificant, the interaction term is positive. This implies that, in contrast to a story where firm liquidity is an important constraint on firm investment, firms with low liquidity actually drive the underlying interest rate response. One possible ex-

Table 7: Heterogeneous Responses to Interest Rate Shocks

	Baseline	Fin. Constraint		Fixed Cost	Cash Flow
		Leverage	Liquidity	Spike	Int. Expenses
1-Year Effect	0.017*** (0.004)	0.024** (0.007)	0.006 (0.005)	0.011*** (0.003)	0.037*** (0.007)
x Characteristic		-0.009 (0.008)	0.016 (0.013)	0.019 (0.019)	-0.024*** (0.006)
Observations	197293	159749	163112	173520	131991
R ²	0.222	0.237	0.231	0.242	0.248

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

planation is that these estimates reflect the endogenous nature of firms' liquidity choices. In particular, if firms accumulate liquidity exactly because they have few profitable investment opportunities, this might also lead these firms to be *less* responsive to changing interest rates. Note that these results do not suggest that liquidity and leverage have no effect on firm investment. Rather, these estimates suggest that they do not mediate the responsiveness of firms to changes in the cost of capital induced by movements in interest rates. However, they may matter for understanding general equilibrium channels that my empirical strategy differences out.

Column (4) also shows that fixed costs play a limited role in shaping the investment response to monetary shocks. The main coefficient is close to the baseline estimate, and the interaction term is an imprecise zero. Column (5) shows that interest expenses are the only characteristic along which there is important heterogeneity in the investment response. These heterogeneous responses provide broad evidence for the mechanism uncovered by the quantitative model: financial frictions that constrain firms when they cannot fund investment out of their cash flows can explain both the quantitative magnitude of the investment response estimated in Section 4, as well as the differences between the tax rate elasticity and interest rate elasticity.

8 Discussion

8.1 Aggregate Implications

This paper shows that the interest rate elasticity of investment is substantially smaller than implied by leading estimates of the investment response to temporary changes in corporate tax policy. However, the empirical response I document is a partial-equilibrium response, estimated after differencing out general-equilibrium forces. A natural question is whether or not differences in this partial-equilibrium response matter in general equilibrium. To assess the broader implications of the empirical facts documented in this paper, I use my estimates to discipline the tractable, heterogeneous-agent New Keynesian model developed in Bilbiie, Känzig and Surico (2022).³² I base my results on this model because it tractably incorporates two important features: (1) the model features capital used in production and subject to investment frictions, and (2) it generates non-trivial MPCs, which thus allows changes in labor incomes as a result of higher investment to affect aggregate consumption.³³

I abstract from idiosyncratic risk; thus, the model features a single optimizing household and a hand-to-mouth household. I set government transfers to zero, so the hand-to-mouth household consumes only out of its labor income. I also include both price and wage-setting frictions. I set all other parameters to the values in Bilbiie, Känzig and Surico (2022). In the model, there is a single parameter ω which governs the responsiveness of investment to marginal q . I consider a similar experiment to the one in Section 6, where I feed in the empirical path of real rates, holding all other prices fixed, and then calibrate ω to hit the empirical impulse response from this project, as well as the baseline estimate from Zwick and Mahon (2017).³⁴ I then show the effect of a monetary shock in general equilibrium in both calibrations that reduces the real rate by 1 p.p. immediately, and then reverts back to steady-state with a persistence parameter of $\rho = 0.6$.

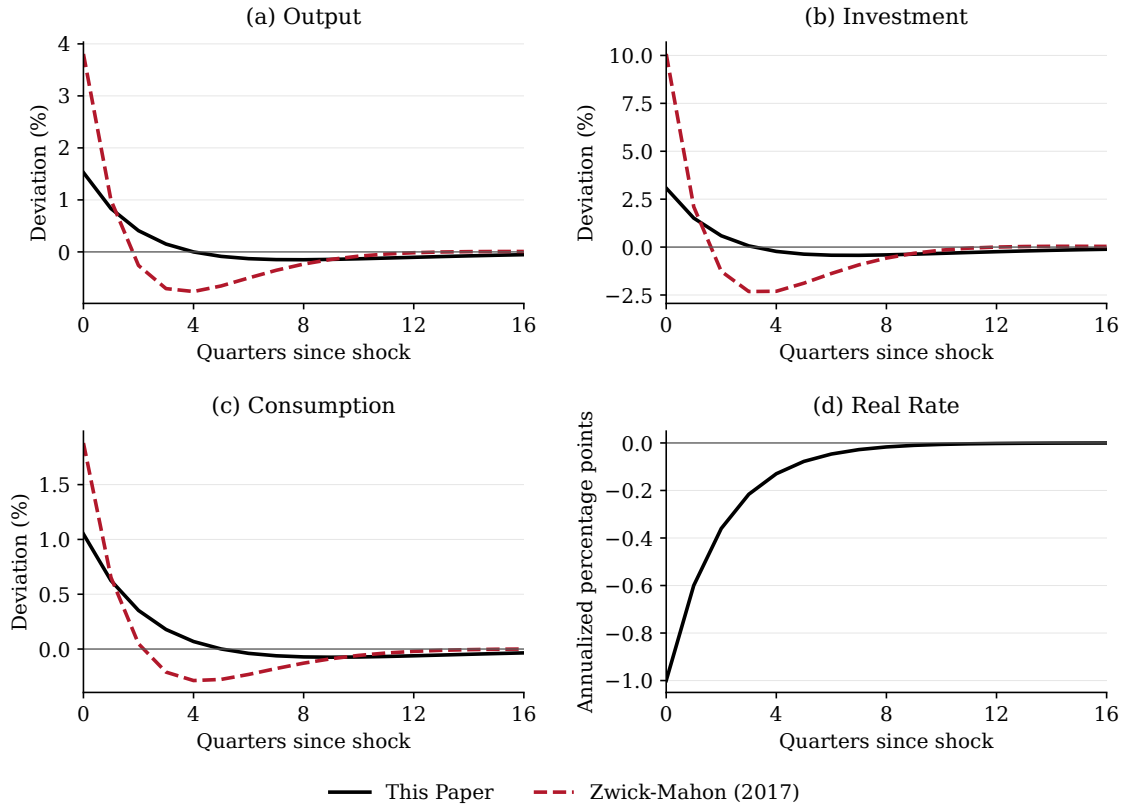
Figure 11 shows that using a smaller investment elasticity substantially weakens the power of monetary policy. Panel (B) shows that a real interest rate cut has significantly smaller effects on investment when the PE investment elasticity is smaller. In the GE model, investment increases by around 3% when calibrated to the investment elasticities in this paper, versus 10% using the Zwick and Mahon (2017) estimates. This translates into smaller aggregate output and consumption effects: the effect of a real rate shock on output is about 60% smaller on

³²A full description of the model is in Appendix B of Bilbiie, Känzig and Surico (2022).

³³In a representative agent version of the model, consumption would be fully pinned down by the representative agent's Euler equation.

³⁴Because there is only one parameter, I calibrate ω so that the one-year PE capital response along the empirical real-rate path exactly matches the 2% estimate from the data (or, alternatively, the implied response from Zwick and Mahon (2017)).

Figure 11: Aggregate Effects of Monetary Policy with Different Investment Elasticities



Notes: This figure plots the effect of a monetary shock in both calibrated versions of the model in Bilbiie, Känzig and Surico (2022). Panels (A) - (C) show the effect on the main outcomes: output, investment, and consumption, respectively. Panel (D) shows the effect on the real interest rate of the policy experiments. Black lines show effects when the model is calibrated to the moments in this paper, while red lines show effects in the model calibrated to Zwick and Mahon (2017).

impact, and the effect on consumption is about 44% smaller when calibrated to the smaller investment elasticity. Thus, the smaller estimated investment elasticities in partial equilibrium translate into smaller effects on output and consumption. Overall, I find that monetary policy has substantially less power to stimulate output and consumption once calibrated to empirical estimates of the partial equilibrium interest rate elasticity of investment.

8.2 Conclusion

This project makes four primary contributions. First, I use a novel empirical strategy to estimate how investment responds to interest rate changes caused by monetary policy, using high-frequency shocks and quasi-experimental techniques. My empirical strategy isolates the partial equilibrium effect of interest rates on investment due to changes in the cost of capital.

These results provide a newly identified, empirical moment based on monetary shocks that can discipline empirical models of investment (Nakamura and Steinsson, 2018b). When I use these novel estimates as portable moments in a general equilibrium, New Keynesian model, I show they have significant implications for the aggregate effect of monetary policy. Second, I show that these effects are substantially smaller than prominent estimates of the effect of corporate taxes on investment. This poses a puzzle for standard neoclassical models, which suggest that the same parameters should govern the investment response to interest rates and tax rates.

Third, I build a quantitative model of heterogeneous firms making corporate investment decisions to highlight how my estimates of the interest rate elasticity of investment can be used to discipline and discriminate between different macroeconomic models. I find that neoclassical models require strong frictions that slow the investment response to interest rates to generate responses that match the estimated elasticities. In addition, only models with financing constraints generate substantially larger responses to tax policy shocks than to interest rate shocks, closing a significant share of the gap between the two empirical elasticities. Tax policy generates free cash flow for firms by reducing their contemporaneous tax liability. In models without costly external financing, these free cash flows are irrelevant. However, when cash flows inside the firm are valuable for funding investments, these cash flows magnify the neoclassical effect of tax rates on investment, generating larger investment elasticities than those from changes in interest rates. The estimated model yields a marginal propensity to invest of 0.61, providing new evidence on an important parameter for models of investment with firm heterogeneity. Finally, I provide evidence that borrowing constraints, like those in the quantitative model, have significant effects on the estimated elasticity consistent with the quantitative model. Firms more likely to be constrained have much smaller investment responses than those less likely to be constrained.

My research has several significant implications for the design of public policy. First, a rigorous understanding of monetary policy requires knowledge of how monetary policy transmits to the real economy. This study adds crucial empirical evidence that changes in interest rates lead to substantive changes in investment, validating a key channel of monetary transmission. However, the estimated elasticities are significantly smaller than implied by the investment response to tax rates, suggesting that the investment response is more limited than would be implied by the existing estimates. In a general equilibrium model calibrated to my estimates, I show that monetary policy has about half the effect on output and consumption as in a more standard calibration.

Second, I provide novel evidence for the importance of a particular type of financial friction that slows the investment response to interest rate changes. This contrasts with other financial frictions prominent in the literature, such as the financial accelerator channel (Bernanke,

Gertler and Gilchrist, 1999), that amplify the investment response to interest rate changes. Third, I present new evidence indicating that countercyclical corporate tax policy can be more effective at stimulating investment than interest rate adjustments. This finding implies that corporate tax changes could serve as a powerful substitute for monetary policy when interest rates are at the zero lower bound or otherwise constrained (Wolf, 2025). Moreover, policy rules that make corporate tax rates countercyclical may act as especially effective automatic stabilizers (McKay and Reis, 2021).

Fourth, my estimates inform the aggregate effects of a broader set of macroeconomic policies, including changes in taxes and tariffs. The interest rate elasticity of investment I estimate is lower than many values in the prior literature and those used by many policy institutions. Applying this lower elasticity would suggest that deficit-financed tax cuts would generate less crowd-out of private investment through higher interest rates than previously assumed. Likewise, the contractionary effects of tariffs would be only modestly offset by any crowd-in of investment from increased tariff revenue.

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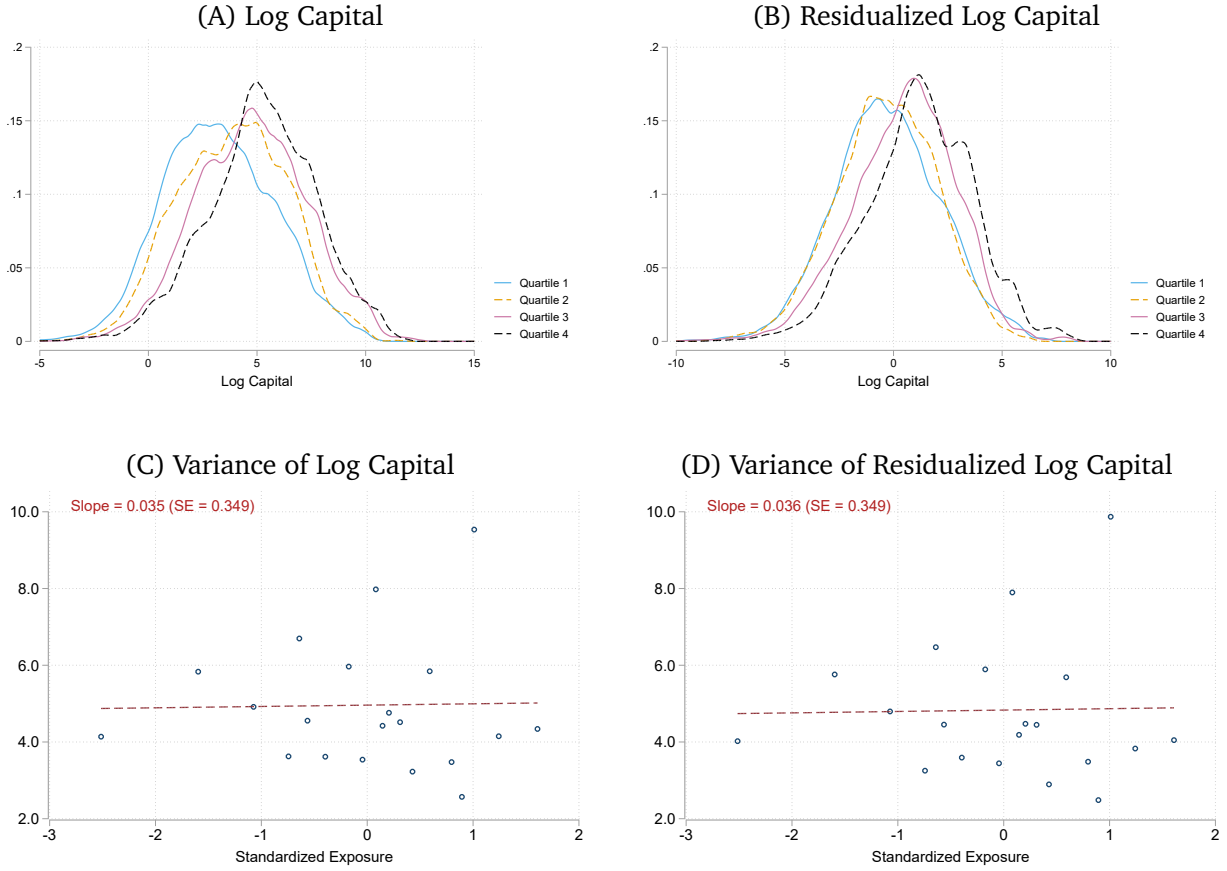
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Appendices Contents

- Appendix A: Additional Figures
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- Appendix D: General Equilibrium New Keynesian Model

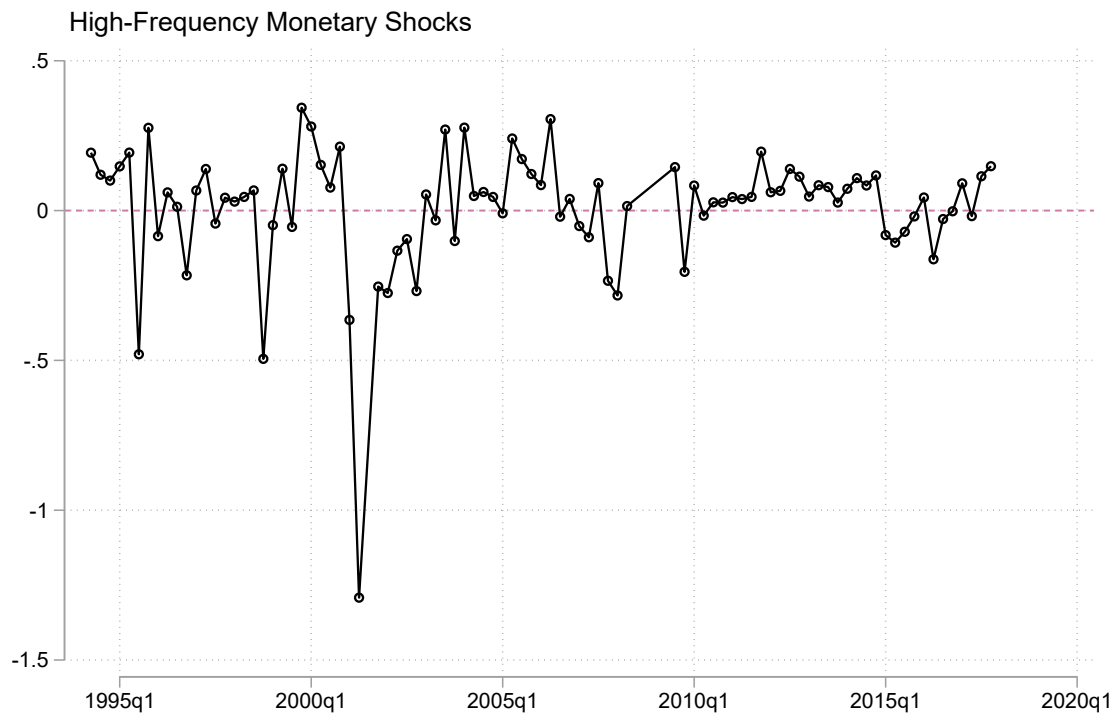
A Additional Figures

Figure A1: Firm-Size Distributions and Variance by Standardized Exposure



Notes: Panels (A) and (B) plot kernel density estimates of log capital and residualized log capital, separately for firms in different quantiles of the residualized depreciation distribution. The log capital distribution plotted in panel (A) is truncated to lie between $[-5, 15]$, while the residualized log capital distribution plotted in panel (B) is truncated to lie between $[-10, 10]$, in order to enhance readability. Panels (C) and (D) plot binned scatter plots of industry-level firm-size variances (as measured by total log capital) against standardized exposure.

Figure A2: Time Series of Monetary Shocks



Notes: This figure shows the time series of high-frequency monetary shocks used in this paper.

Figure A3: The Effect of Monetary Shocks on Firm Financing

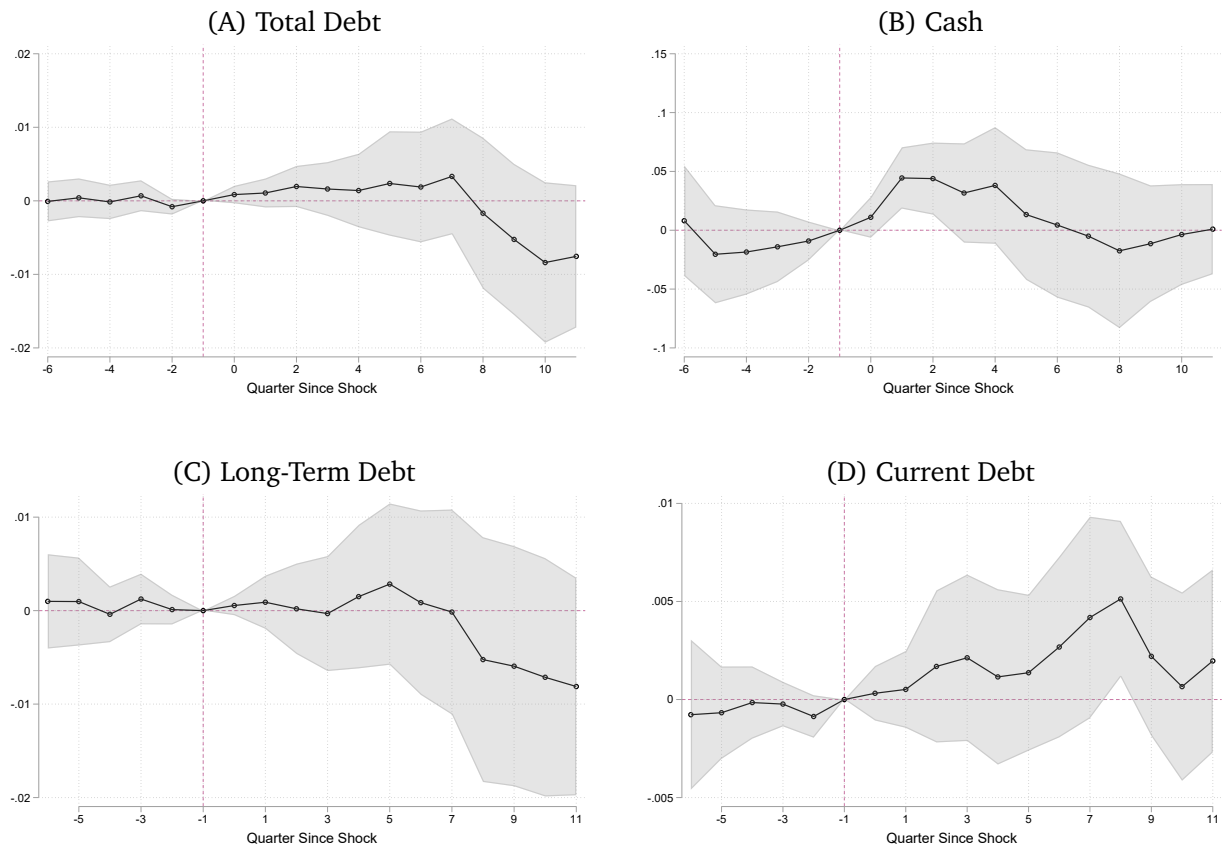
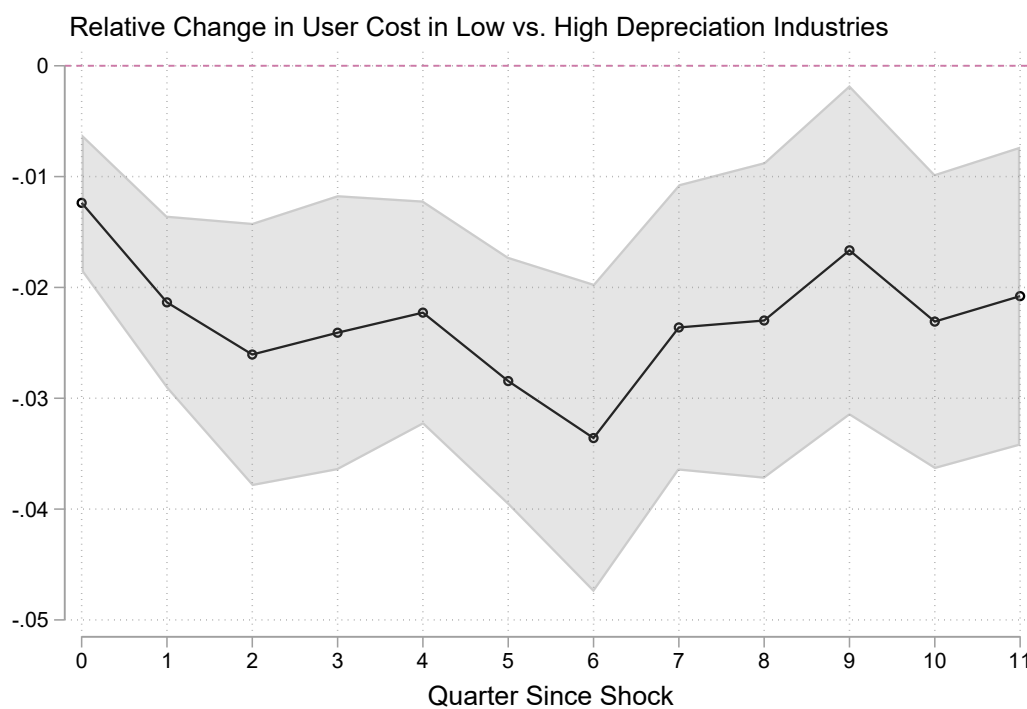
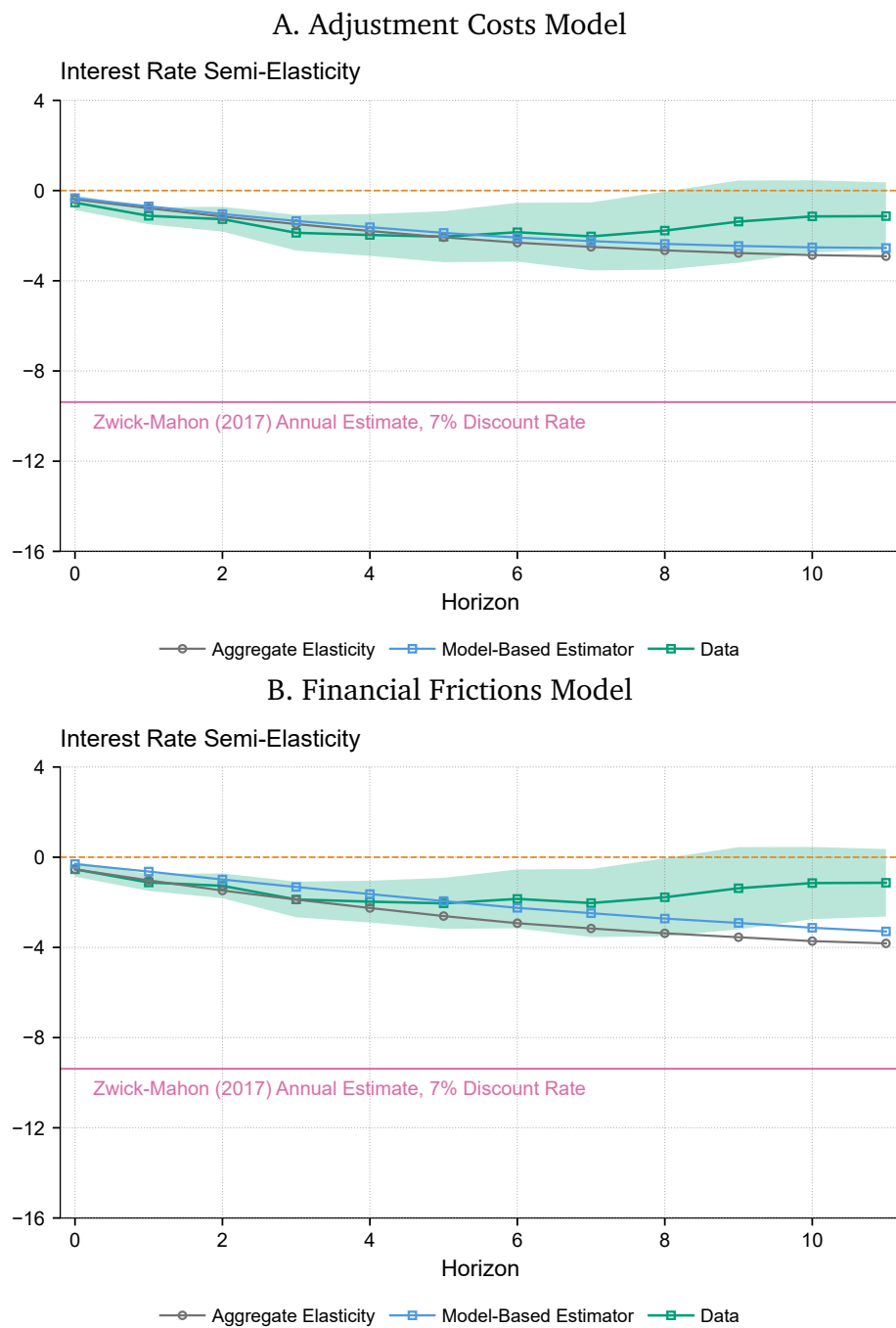


Figure A4: Firm-Level Impulse Response: Effect on Log User Cost of Capital



Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks interacted with the depreciation rate. The outcome variable is the change in the log user cost. The regression includes sector-by-time fixed effects. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

Figure A5: Validation of the Model-Based Interest Rate Elasticity



Notes: This figure compares the aggregate interest rate elasticity with the low-versus-high-depreciation estimator.

B Appendix Tables

Table A1: Residualized Depreciation Rates

Industry	Residualized Depreciation
Education Services	-4.2%
Primary Metal Manufacturing	-4.0%
Rail Transportation	-3.9%
Other Services	-3.4%
Accommodation	-3.4%
Broadcasting	-3.1%
Other Transportation Equipment	3.0%
Computer and Electronic Products	3.6%
Misc. Professional Services	5.3%
Car Manufacturing	6.6%
Truck Transportation	8.7%
Computer Systems Design	10.3%

Notes: This table shows the highest and lowest depreciation industries after residualizing the depreciation rates on major-industry fixed effects.

Table A2: Heterogeneous Responses to Interest Rate Shocks by Firm Size

	1-Year Effect		2-Year Effect	
Baseline Coefficient	0.017*** (0.004)	0.022* (0.009)	0.019* (0.007)	0.038 (0.022)
x Quintile 2		-0.006 (0.011)		-0.047 (0.026)
x Quintile 3		-0.011 (0.011)		-0.018 (0.024)
x Quintile 4		-0.007 (0.012)		-0.020 (0.019)
x Quintile 5		0.002 (0.007)		-0.007 (0.016)
Observations	197293	196877	182076	181682
R ²	0.222	0.223	0.316	0.317

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A3: Heterogeneous Responses to Interest Rate Shocks by Firm Size

	1-Year Effect			2-Year Effect		
	All Firms	Small Firms	Big Firms	All Firms	Small Firms	Big Firms
Main Coefficient	0.017*** (0.004)	0.016** (0.005)	0.016** (0.005)	0.019* (0.007)	0.009 (0.006)	0.024* (0.009)
Observations	197293	106151	90407	182076	98454	82953
R ²	0.222	0.214	0.250	0.316	0.300	0.358

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A4: The Effect of Monetary Shocks on Investment Prices

	L4. Ln(P)	L2.Ln(P)	Ln(P)	F1. Ln(P)	F2. Ln(P)	F4. Ln(P)	F8. Ln(P)
Exposure x Shock	-0.099 (0.077)	-0.077 (0.076)	-0.053 (0.077)	-0.042 (0.077)	-0.032 (0.076)	-0.032 (0.074)	-0.042 (0.067)
<i>N</i>	3132	3204	3240	3204	3168	3096	2952
<i>R</i> ²	0.75	0.74	0.73	0.73	0.73	0.74	0.74
Capital FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date-Class FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

C Robustness Appendix

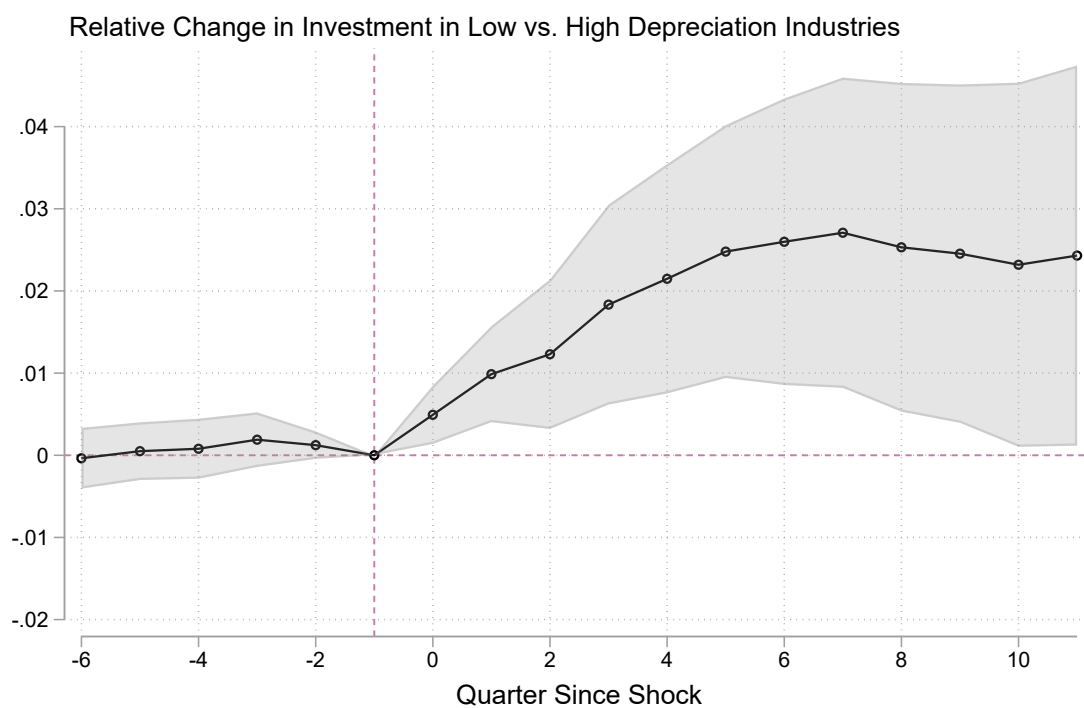
Appendix Figure A6 re-estimates the baseline specification, using cumulated capital expenditures normalized by the lagged capital stock as the dependent variable. Despite using a completely different variable for investment, the results are quantitatively very similar and statistically significant. Appendix Figure A7 also shows the coefficients are essentially unchanged when weighting by the lagged capital stock.

Appendix Figures A8, A9, and A10 show that the baseline results are not sensitive to the choice of controls used in estimating the impulse response. The results are qualitatively unchanged when dropping the baseline controls, allowing for more flexible interactions between the baseline controls, or adding additional controls for firm size or estimates of investment irreversibility interacted with the monetary shock (Kermani and Ma, 2023).³⁵ Appendix Figure A11 also shows that the results are robust to excluding periods with significant changes to other policies that might differentially affect industries with different types of capital goods. Appendix Figure A11, Panel (A) shows robustness to dropping periods around significant announcements related to Bonus Depreciation. Appendix Figure A11, Panel (B) shows that quantitative easing does not confound the baseline estimates (Hubert de Fraisse, 2024).

Appendix Figures A12 and A13 show that the baseline results are minimally affected by changes in risk that might occur due to Fed announcements. The results are similar when allowing the response of investment to changes in the term premia to vary with the depreciation rate, or for flexible interactions of stock price movements that occur around monetary policy announcements with firm-level beta. Appendix Figure A14 shows robustness tests to alternative ways of constructing the exposure variable. Finally, Appendix Figures A15 and A16 show that the results are quantitatively similar (if less precisely estimated) when excluding shocks in the major easing cycles of the 2000s or when using alternative shocks to interest rates.

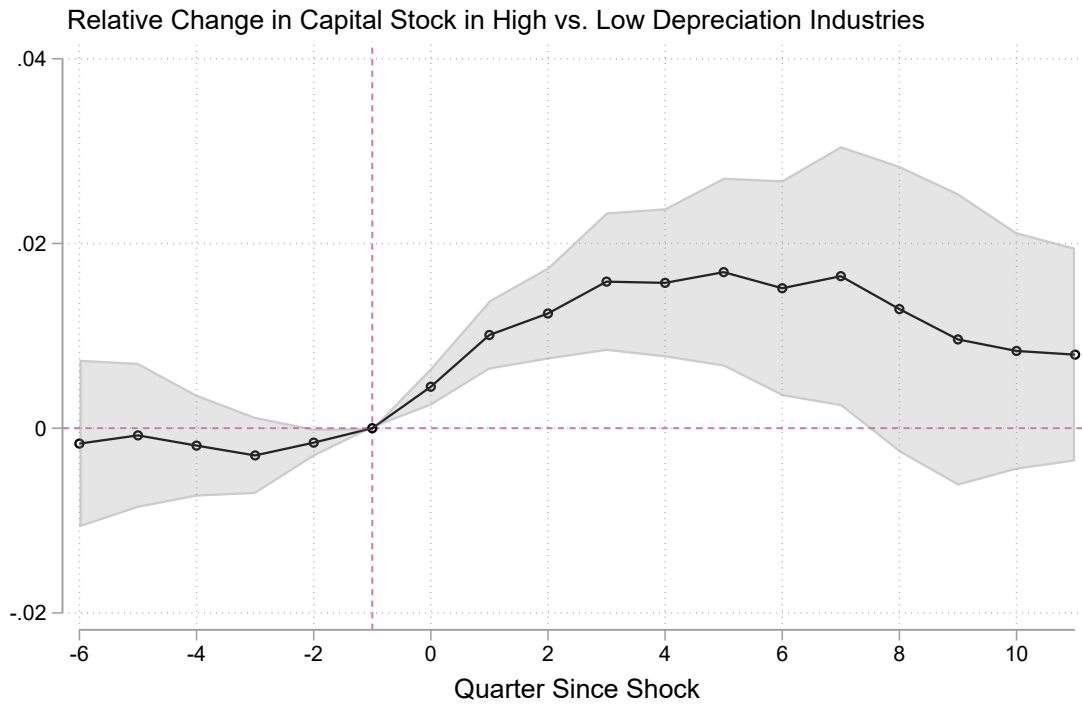
³⁵Note that not all industries have sufficient data on recovery rates, and thus the specification with the recovery rate control includes fewer observations.

Figure A6: Firm-Level Impulse Response: Using Cumulated Investment As Dependent Variable



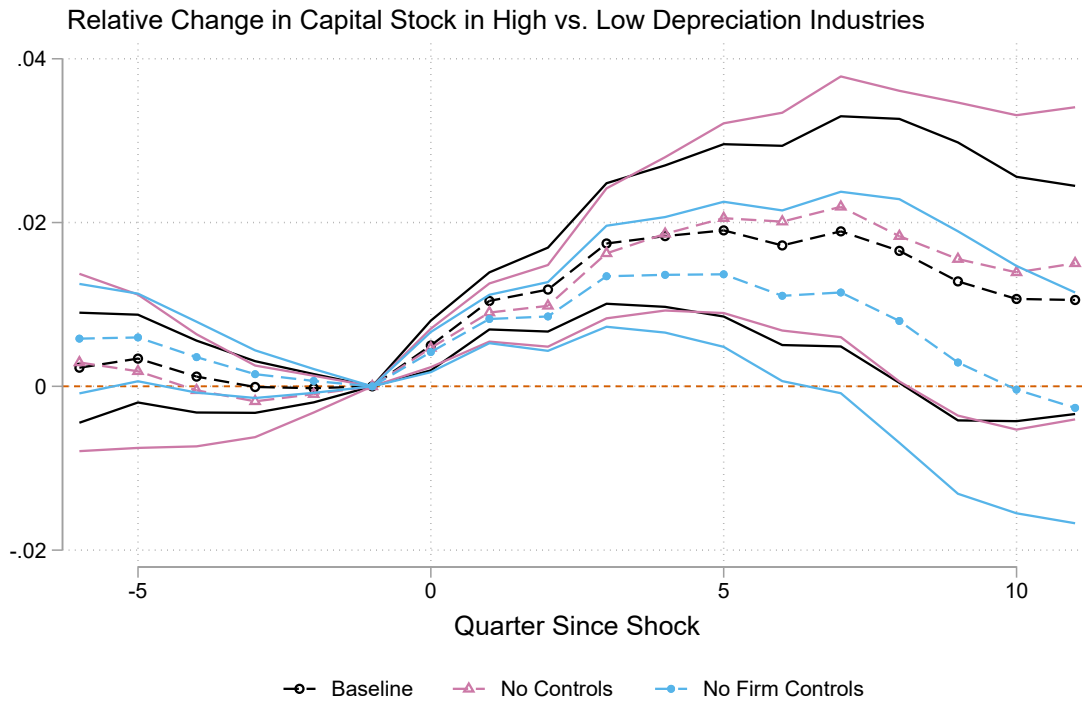
Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level outcomes. The outcome variable is cumulated gross investment, normalized by the lagged capital stock. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

Figure A7: Firm-Level Impulse Response: Weighted Regression



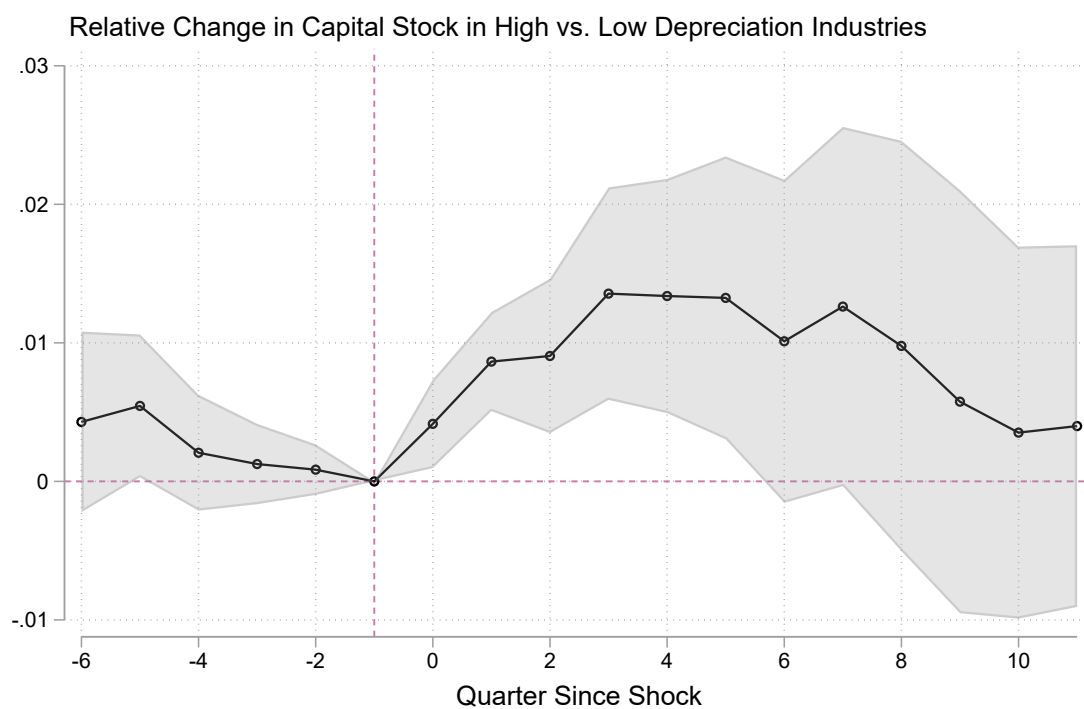
Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level outcomes. The outcome variable is net investment, and the regression is weighted by the normalized (within-quarter) lagged capital stock. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

Figure A8: Firm-Level Impulse Response: Robustness to Control Set



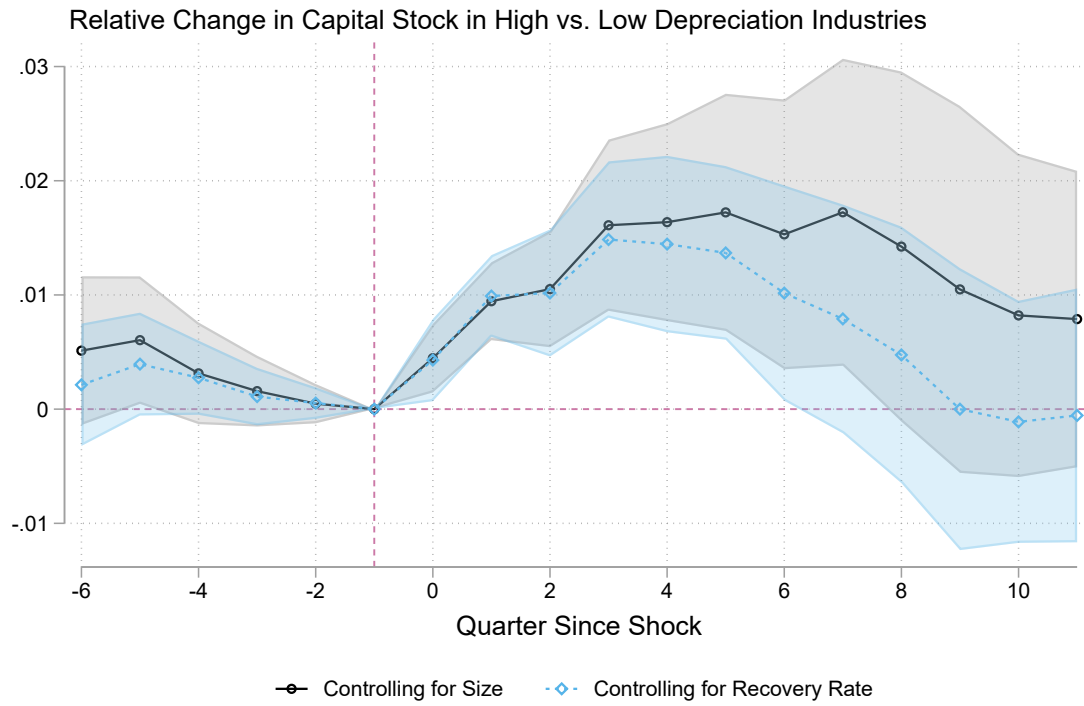
Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level outcomes. The outcome variable is the change in net investment. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

Figure A9: Firm-Level Impulse Response: Robustness to Control Set



Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level outcomes. The outcome variable is the change in net investment. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

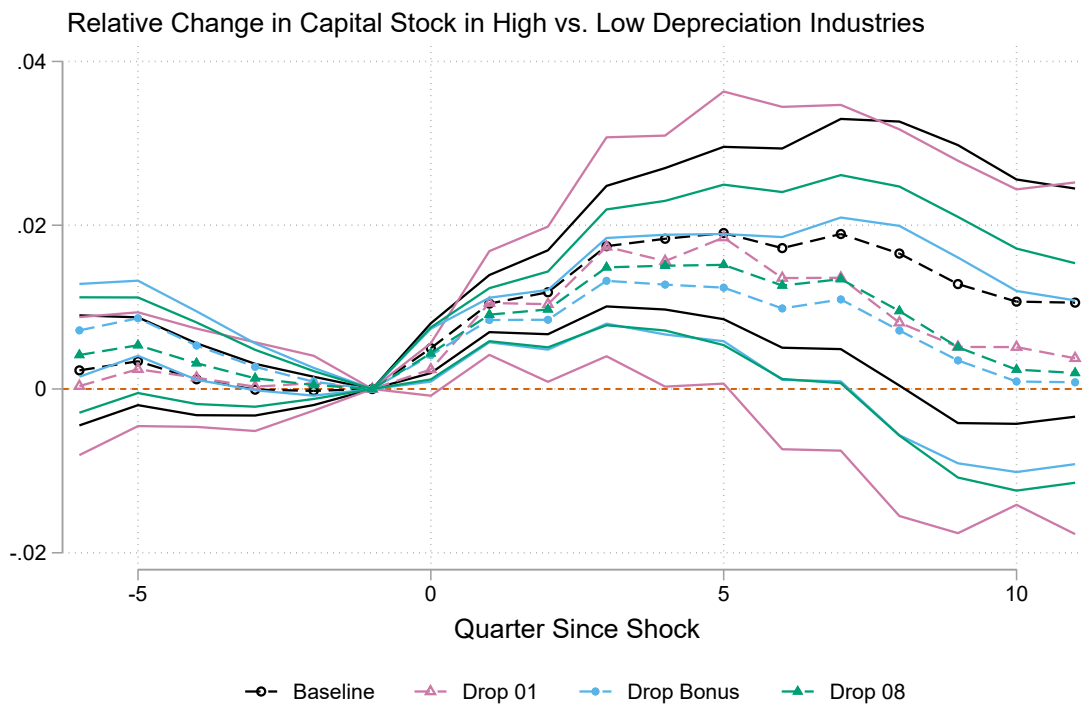
Figure A10: Firm-Level Impulse Response: Robustness to Control Set



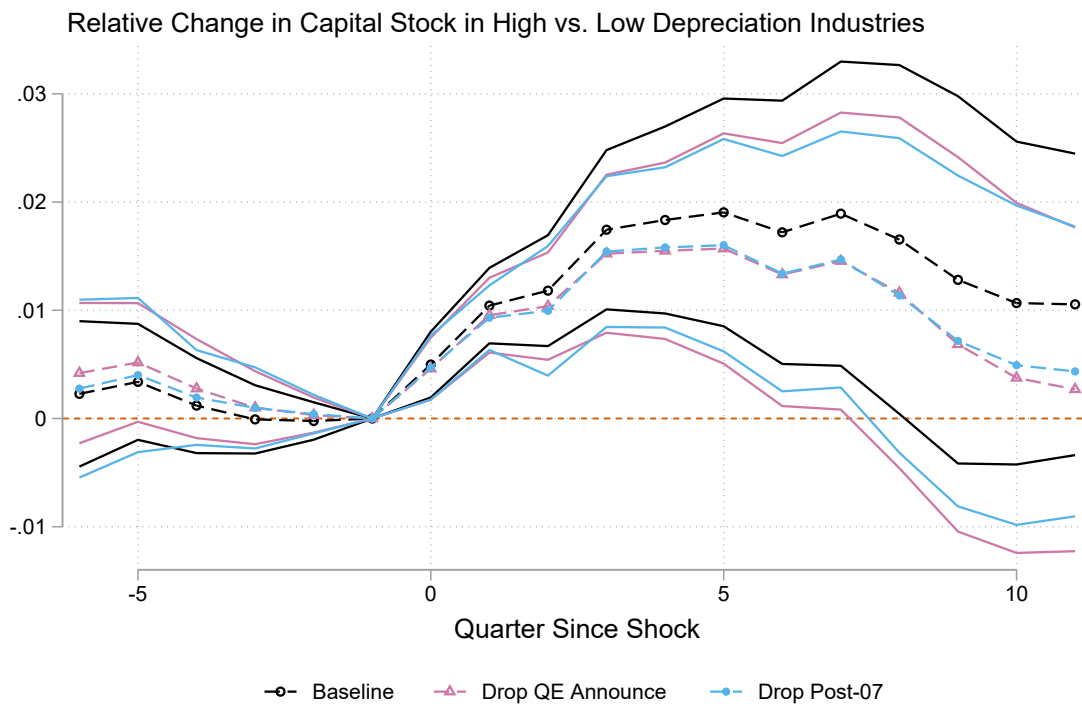
Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level outcomes. The outcome variable is the change in net investment. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

Figure A11: Firm-Level Impulse Response: Robustness to Restricting Time Dimension

A. Dropping Periods Around Bonus Announcements

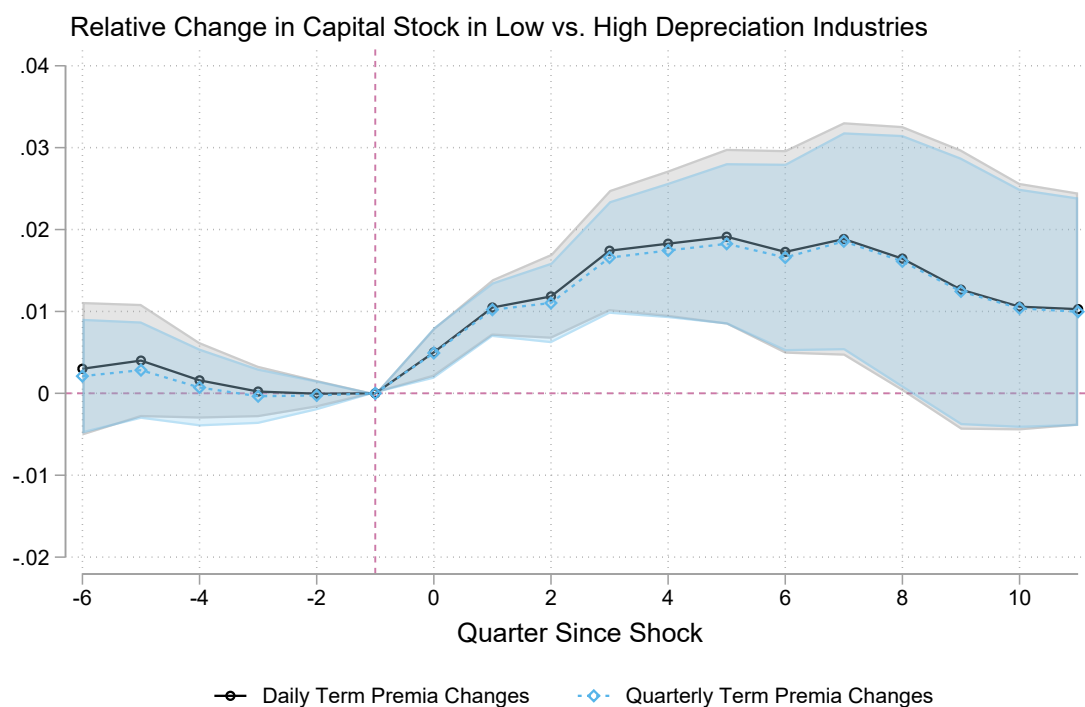


B. Dropping Periods Around QE Announcements



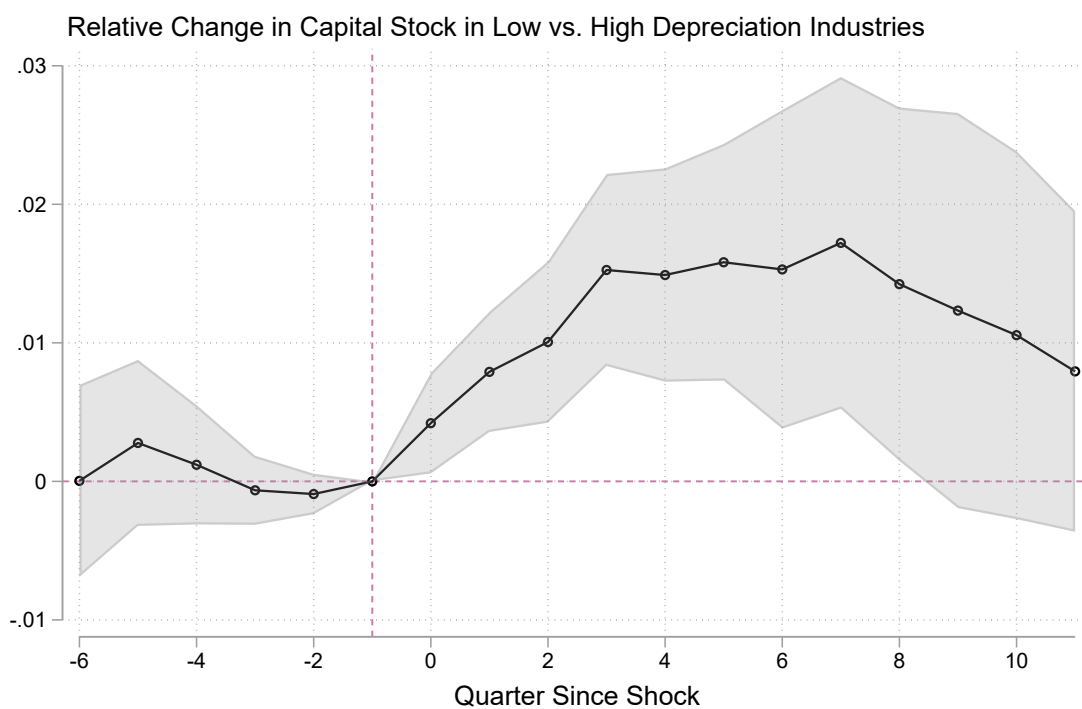
Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level outcomes. Panel (A) shows impulse responses excluding quarters around the announcement of the first two rounds of bonus depreciation. Panel (B) shows impulse responses excluding quarters around the announcement of QE policies. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

Figure A12: Firm-Level Impulse Response: Allowing for Flexible Term-Premia Interactions



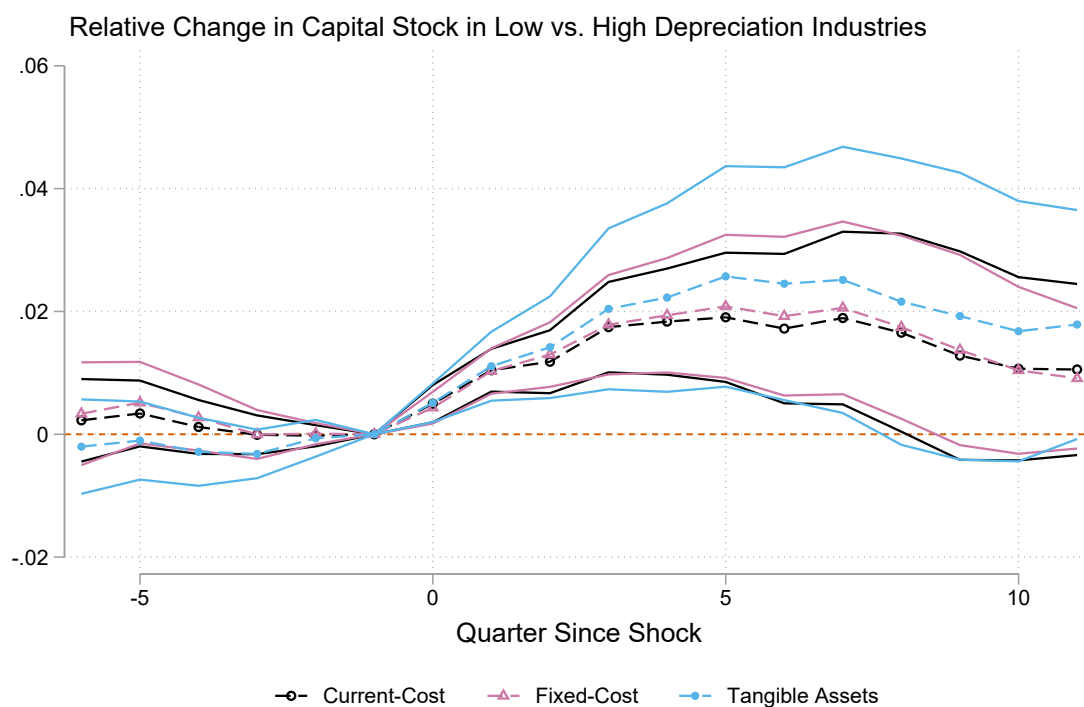
Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level outcomes. The outcome variable is the change in net investment. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level. The impulse response in black shows the results controlling for daily changes in the term premia on Fed announcement days interacted with the depreciation rate, while the impulse response in blue controls for overall, quarterly changes in the term premia interacted with the depreciation rate. Estimates from the term premium come from Adrian, Crump and Moench (2013)

Figure A13: Firm-Level Impulse Response: Allowing for Flexible Firm-Level Beta Controls



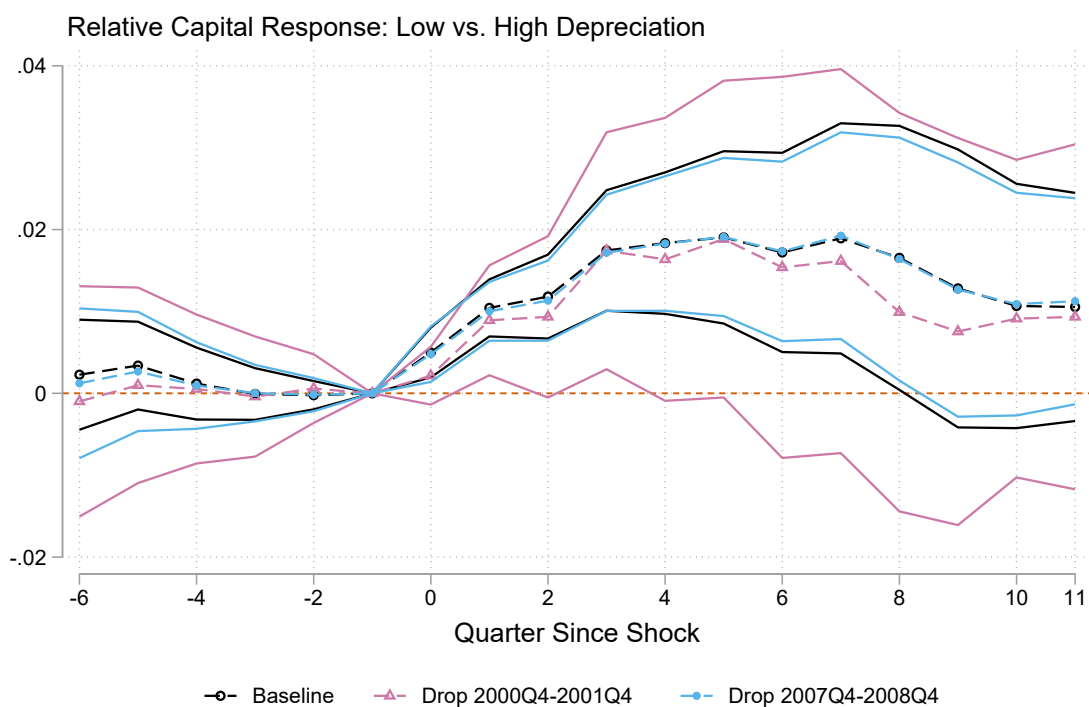
Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level outcomes. The outcome variable is the change in net investment. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level.

Figure A14: Firm-Level Impulse Response: Robustness to Definition of the Exposure Variable



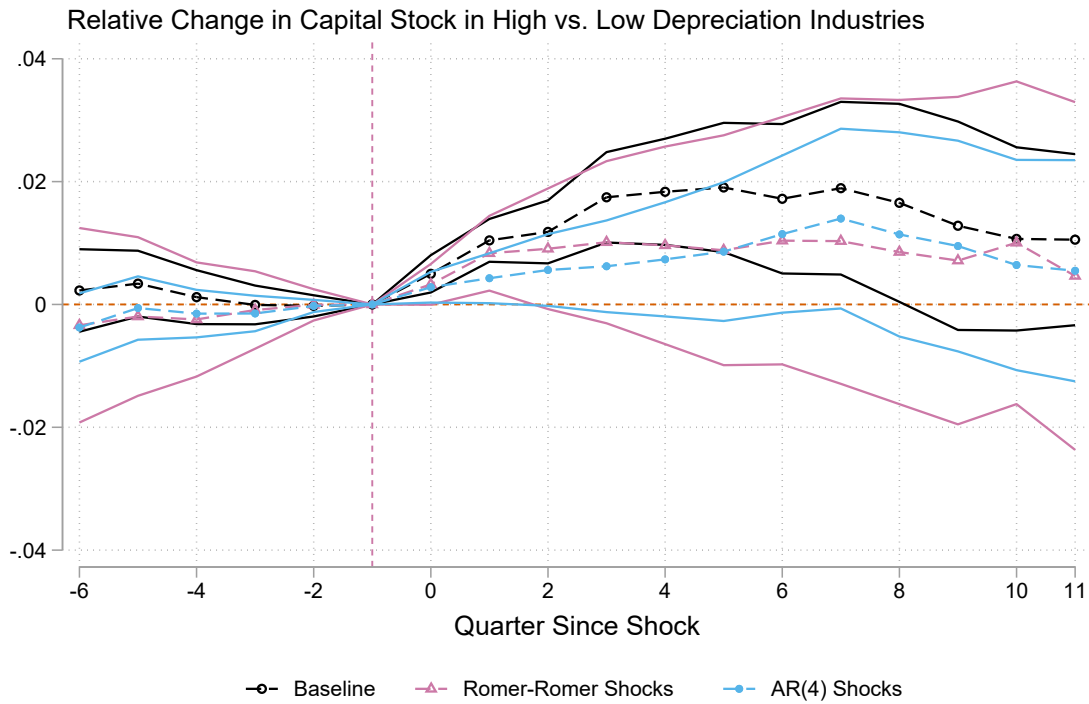
Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level investment. The outcome variable is the change in net investment. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level. The line in black shows the baseline exposure variable, where exposure is computed using the current-cost capital stock. The line in pink shows the estimates when exposure is computed using the fixed-cost capital stock, and the line in blue shows the results when the exposure variable is computed using tangible assets only.

Figure A15: Firm-Level Impulse Response: Excluding the 2001 and 2008 Windows



Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level investment. The outcome variable is the change in net investment. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level. The line in black shows the baseline estimate, while the line in pink drops the early 2000s easing cycle and the line in blue drops the easing cycle during the Great Recession.

Figure A16: Firm-Level Impulse Response: Alternative Interest Rate Shocks



Notes: This figure plots the Jordà (2005)-style impulse response of the effect of monetary policy shocks on firm-level investment. The outcome variable is the change in net investment. 95% confidence intervals are plotted, and standard errors are two-way clustered at the industry and date level. The line in black shows the baseline estimate, while the line in pink uses the narrative shocks from Romer and Romer (2004) and the line in blue uses residuals from an AR(4) process for the real rate. Both alternative shocks are scaled to equal the same peak effect on the real rate as the baseline estimates, and thus are directly comparable.

D General Equilibrium New Keynesian Model

This section integrates the partial equilibrium firm block from Section 5. The New Keynesian closure closely follows the models in Ottonello and Winberry (2020) and Fang and Zhou (2025). The heterogeneous firm problem is nearly identical to the baseline model. The one difference is that in GE, the household discounts firms' continuation value not using the real interest rate directly, but rather using the household's stochastic discount factor Δ_t . Thus, the firm problem is:

$$V_t(k, z, B, \delta) = \max_{k', n} p_t f(k, z, n_t) - w_t n - q_t * (k' - (1 - \delta) * k) - AC(k', k, I_t; \zeta) - T(\pi_t, \tau_t, b_t, B, I_t) - \Phi(d_t^0, \pi_t, p_t, Y_t) + \mathbb{E}_t[\Delta_t V_{t+1}(k', z', B', \delta)] \quad (10)$$

D.1 New Keynesian Block

Heterogeneous firms produce a series of identical goods, indexed by the unit interval, and sell them (at a price that can perfectly adjust) to a mass of retailers also indexed by the unit interval. These retailers take the undifferentiated goods given by the production firms and differentiate them into different varieties using a one-to-one transformation. These retailers then sell their differentiated goods to a final goods firm, which aggregates them up using a CES production function into the final good consumed by the household, generating a demand curve faced by the retailers. Retailers face a Rotemberg price adjustment cost $\frac{\phi}{2} (\frac{\tilde{p}_t}{p_{t-1}^*} - 1)^2 Y_t$, where Y_t is the final good and $\tilde{p}$ is the retailers price. The retailers' profit maximization problem generates a (linearized) New-Keynesian Phillips Curve:

$$\log \Pi_t = \frac{\gamma - 1}{\phi} \log \frac{p_t}{p^*} + \beta \mathbb{E}_t[\log \Pi_{t+1}]$$

Where γ is an elasticity of substitution over intermediate goods, ϕ is a coefficient on the price adjustment cost, and $p^* = \frac{\gamma-1}{\gamma}$ is the steady-state price of output of the heterogeneous production firms.

In this economy, the nominal interest rate is set by the central bank, which follows a Taylor Rule in logs:

$$\log r_t = \log \frac{1}{\beta} + \sigma \log \Pi_t + \epsilon_t^m$$

Where ϵ_t^m is a shock and σ is the weight on inflation in the central bank reaction function.

The representative household maximizes:

$$\mathbb{E}_0 \sum_t \beta^t (\log C_t - \Upsilon L_t)$$

subject to the following budget constraint:

$$P_t C_t + \frac{1}{r_t} B_t \leq B_{t-1} + w_t n_t + D_t$$

where D_t is dividends and B_t is bonds. Household optimization gives the labor-supply decision and the following equation for the SDF:

$$\Delta_t = \frac{\Pi_{t+1}}{r_t}$$

Relative to the partial equilibrium model, there are also two crucial market clearing conditions. In particular, total output Y_t has to equal consumption from the households plus investment from the firms, net of adjustment costs. Labor supplied by the representative household has to equal labor demanded by the intermediate goods firms. Thus, aggregate inflation and p_t are determined by the New Keynesian Phillips Curve, and w_t is the endogenous price from the household's and firms' first-order conditions that ensures the labor market clears.

In the baseline New Keynesian model, I hold q_t fixed, following the evidence in Section 4 that capital goods prices do not respond to monetary shocks. However, one can easily endogenize q_t following Ottonello and Winberry (2020). In particular, we now require a capital goods producer to produce the investment technology. Ottonello and Winberry (2020) use a production technology for capital, which generates the following equilibrium price of capital:

$$q_t = \left(\frac{I_t}{K_t} \right)^{\frac{1}{\xi}}$$

where I^* is the steady-state investment rate and ξ represents the investment supply elasticity. This adds another equilibrium condition, where investment supply from the capital goods firm equals investment demand from the intermediate goods firm. q_t is the price that clears this market.

In this case, an equilibrium takes the initial firm-size distribution $\mu_0(k, z, B, \delta)$ as given, as well as a sequence of policy parameters $\{\tau_t, b_t\}$. Now, an equilibrium of the model is a set of value functions $V_t(k, z, B, \delta)$, policy functions $k_t(z, k, B, \delta)$, and a firm-size distribution $\mu(k, z, B, \delta)$, such that all firms optimize and the distribution of firms is consistent with the decision rules. In addition, $\{p_t, w_t, q_t, \Pi_t, \Delta_t\}$ must satisfy the New Keynesian equilibrium relationships such that households optimize and all markets clear.